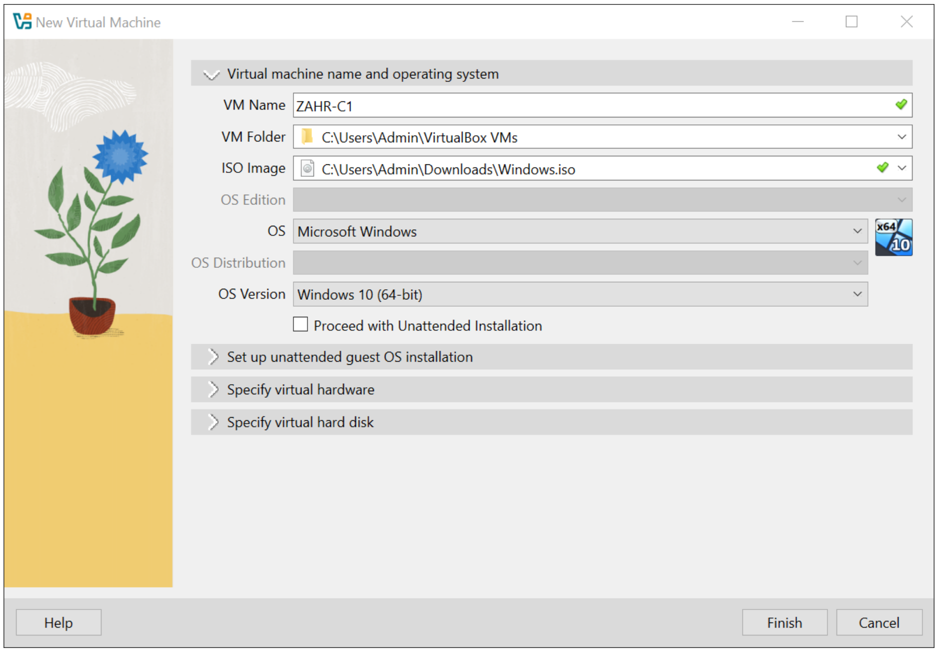
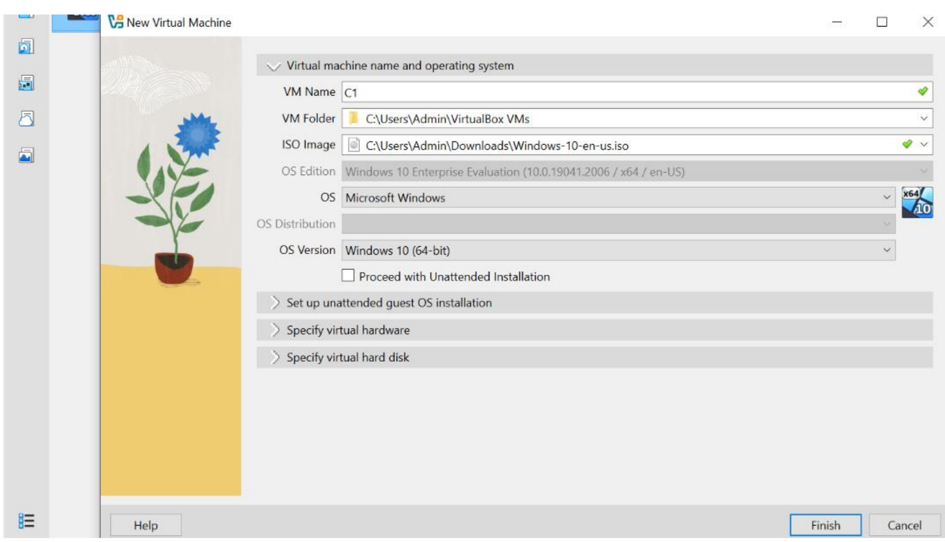
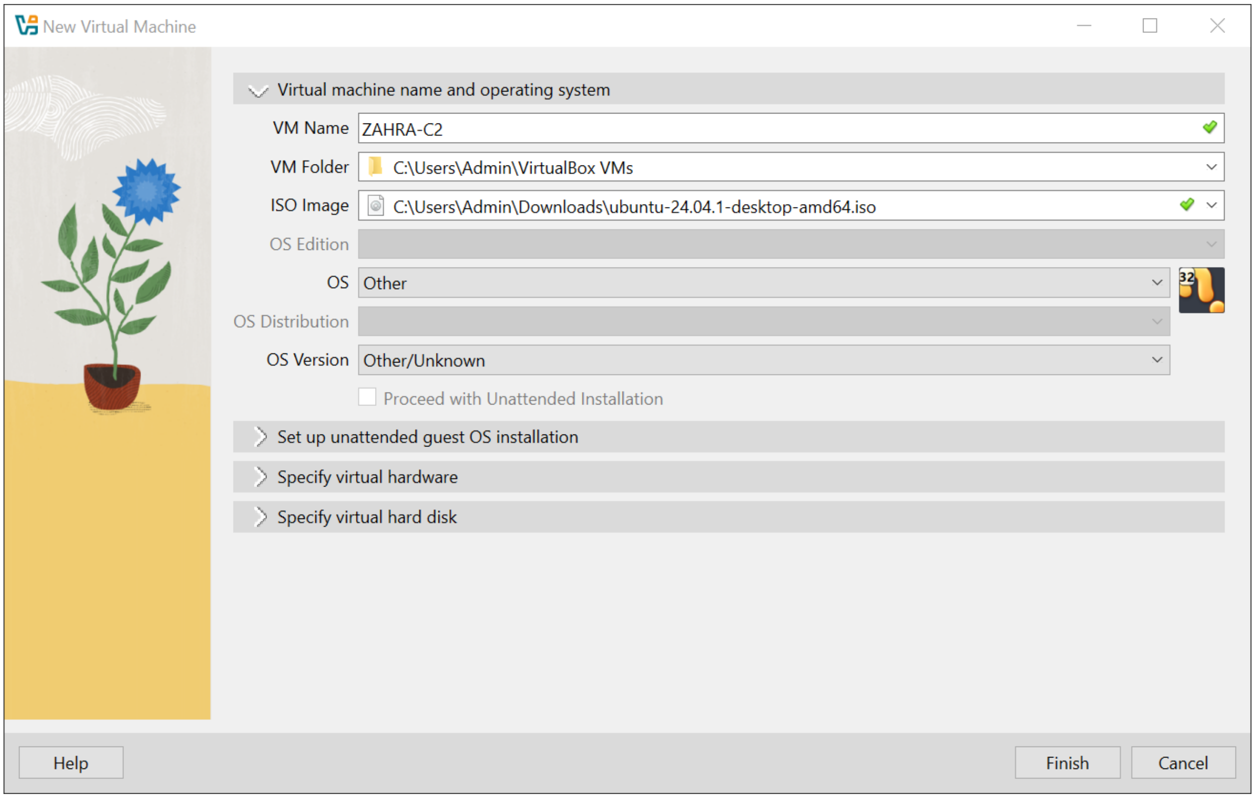
NAME:	Zahra said Aldayri.
ID:	56s1986.

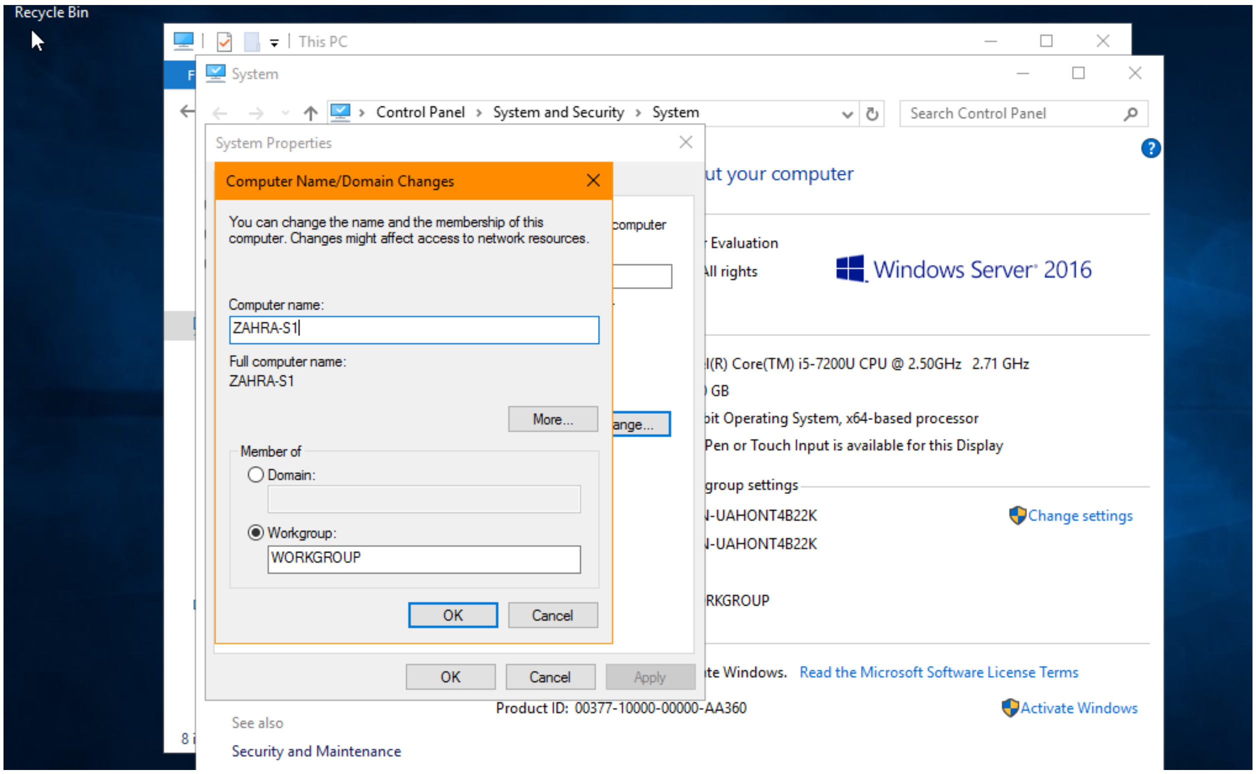
1. Install Windows10, Windows2016, and Ubuntu on Oracle VB.

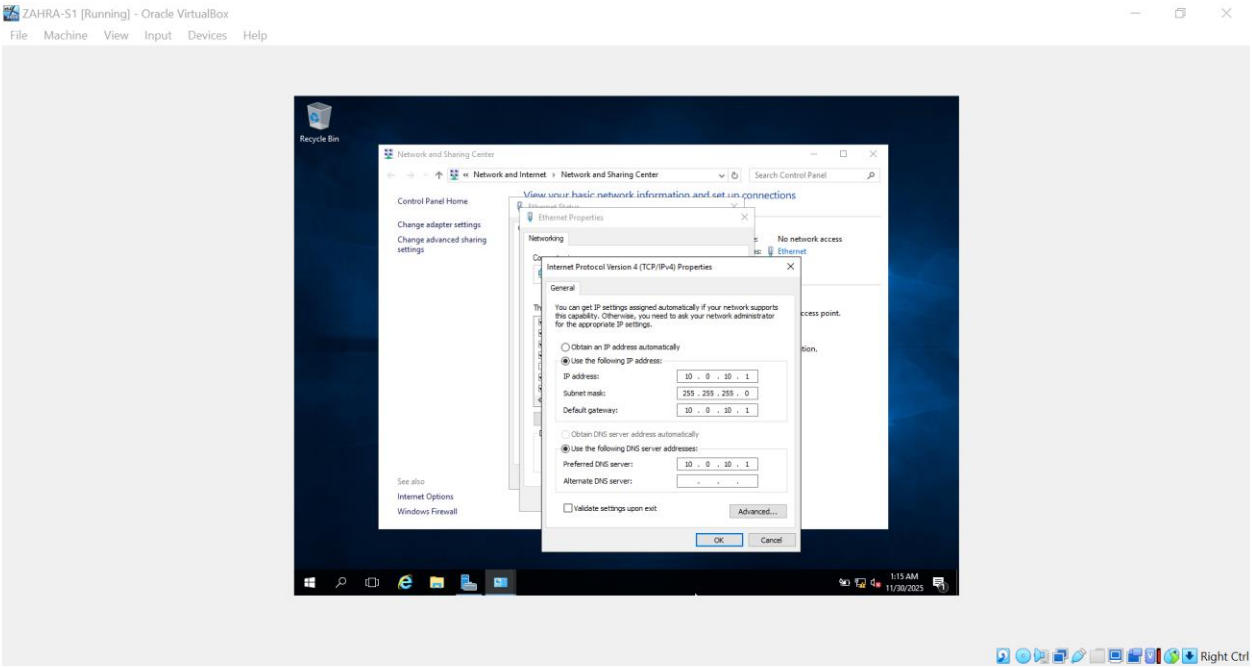


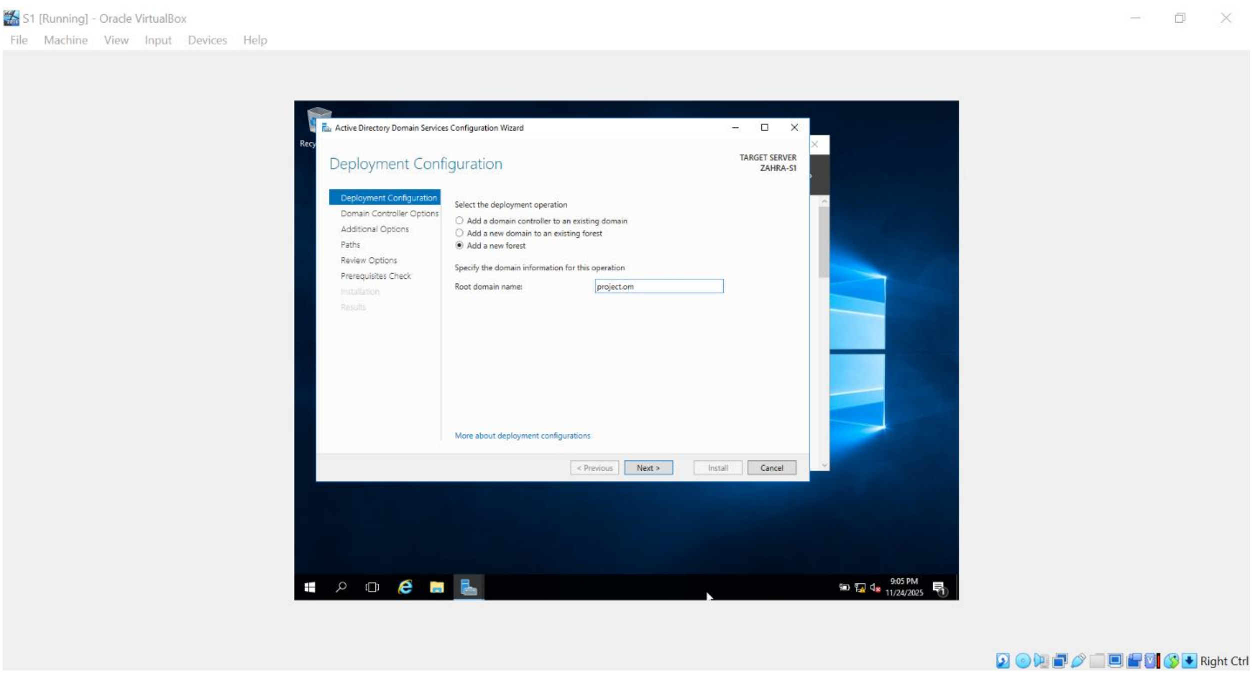


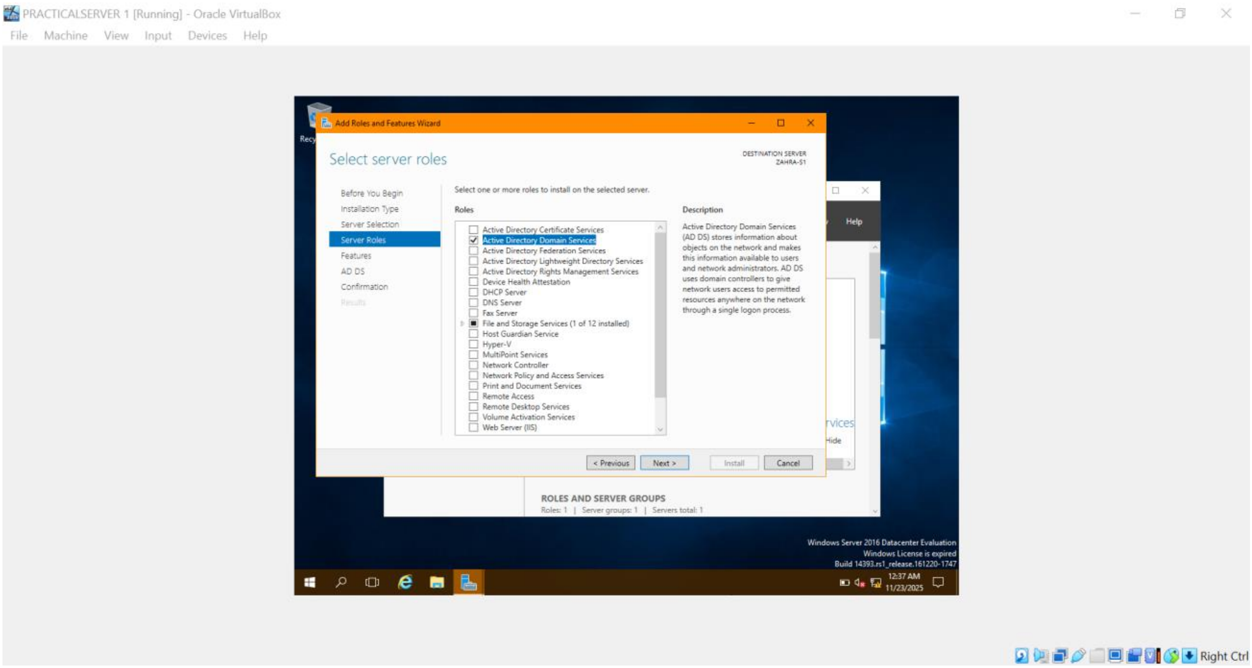

New Virtual Machine
Virtual machine name and operating system
VM Name
ZAHRA-C2
VM Folder
C:\Users\Admin\VirtualBox VMs
ISO Image
C:\Users\Admin\Downloads\ubuntu-24.04.1-desktop-amd64.iso
OS Edition
OS
Other
OS Distribution
OS Version
Other/Unknown
Proceed with Unattended Installation
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Help
Finish
Cancel

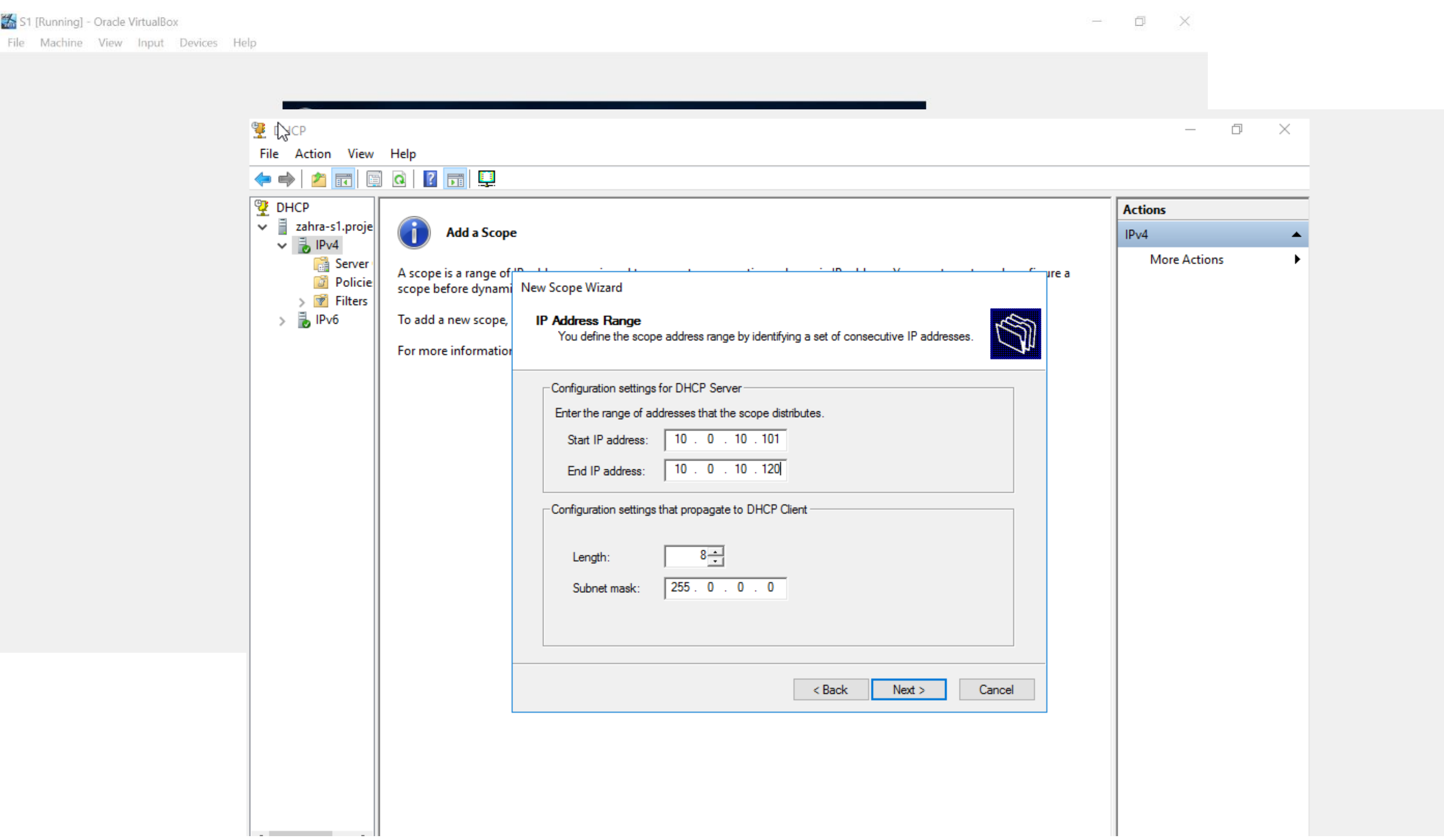
2. **Configure Windows 2016 as YourName-S-1 with proper ip address, enable & configure services like Active Directory, DNS, DHCP etc. Server Address: 10.0.10.1, Client Range: 10.0.1.101–10.0.10.120**

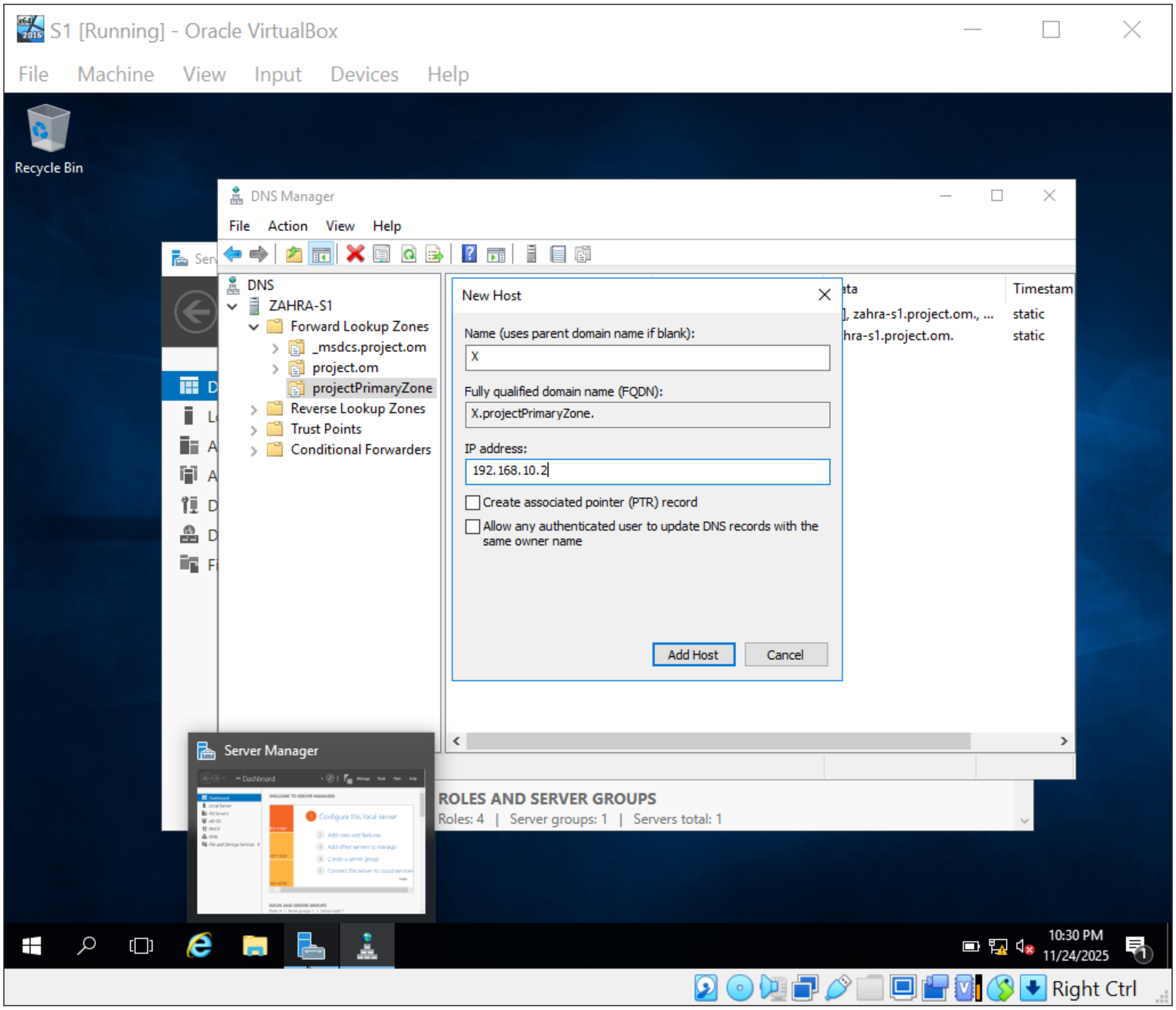


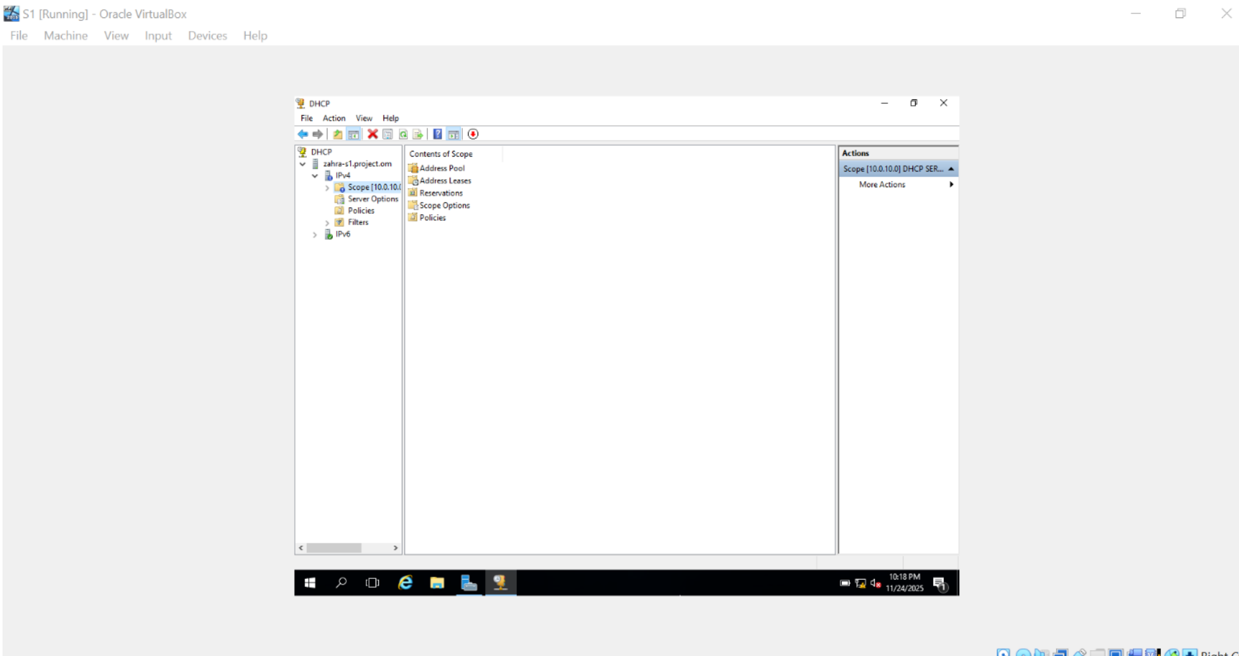

ZAHRA-S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Network and Sharing Center
Internet Protocol Version 4 (TCP/IPv4) Properties
Use the following IP address:
IP address: 10 . 0 . 10 . 1
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 10 . 0 . 10 . 1
Use the following DNS server addresses:
Preferred DNS server: 10 . 0 . 10 . 1
OK
Cancel

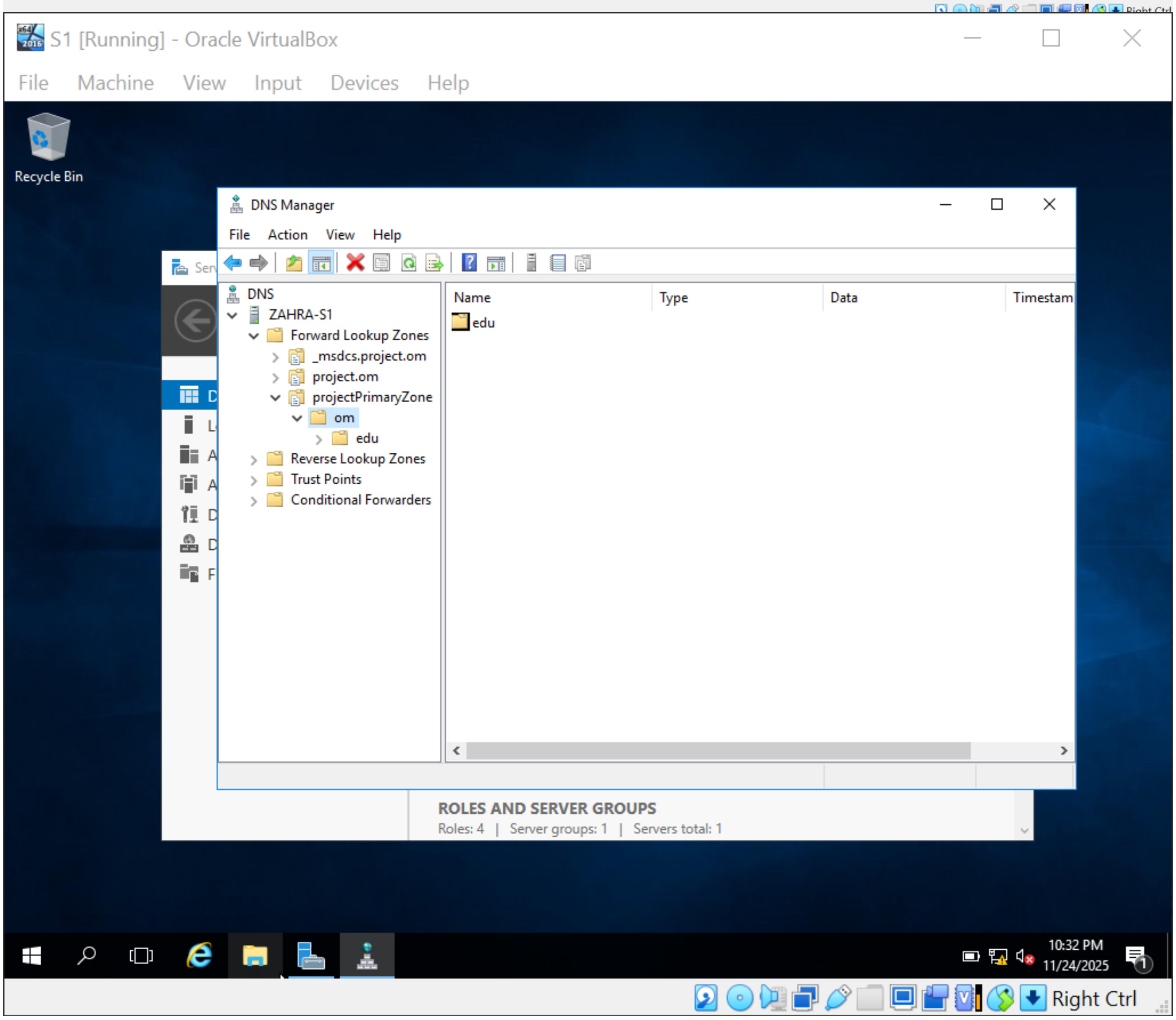

S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Active Directory Domain Services Configuration Wizard
Deployment Configuration
TARGET SERVER
ZAHRA-S1
Select the deployment operation
Add a domain controller to an existing domain
Add a new domain to an existing forest
Add a new forest
Specify the domain information for this operation
Root domain name:
project.om
More about deployment configurations
< Previous
Next >
Install
Cancel


PRACTICALSERVER 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Add Roles and Features Wizard
Select server roles
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
Confirmation
Results
Select one or more roles to install on the selected server.
Roles
Description
Active Directory Certificate Services
Active Directory Domain Services
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server
Fax Server
File and Storage Services (1 of 12 installed)
Host Guardian Service
Hyper-V
MultiPoint Services
Network Controller
Network Policy and Access Services
Print and Document Services
Remote Access
Remote Desktop Services
Volume Activation Services
Web Server (IIS)
Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.
< Previous
Next >
Install
Cancel
ROLES AND SERVER GROUPS
Windows Server 2016 Datacenter Evaluation
Windows License is expired
Right Ctrl

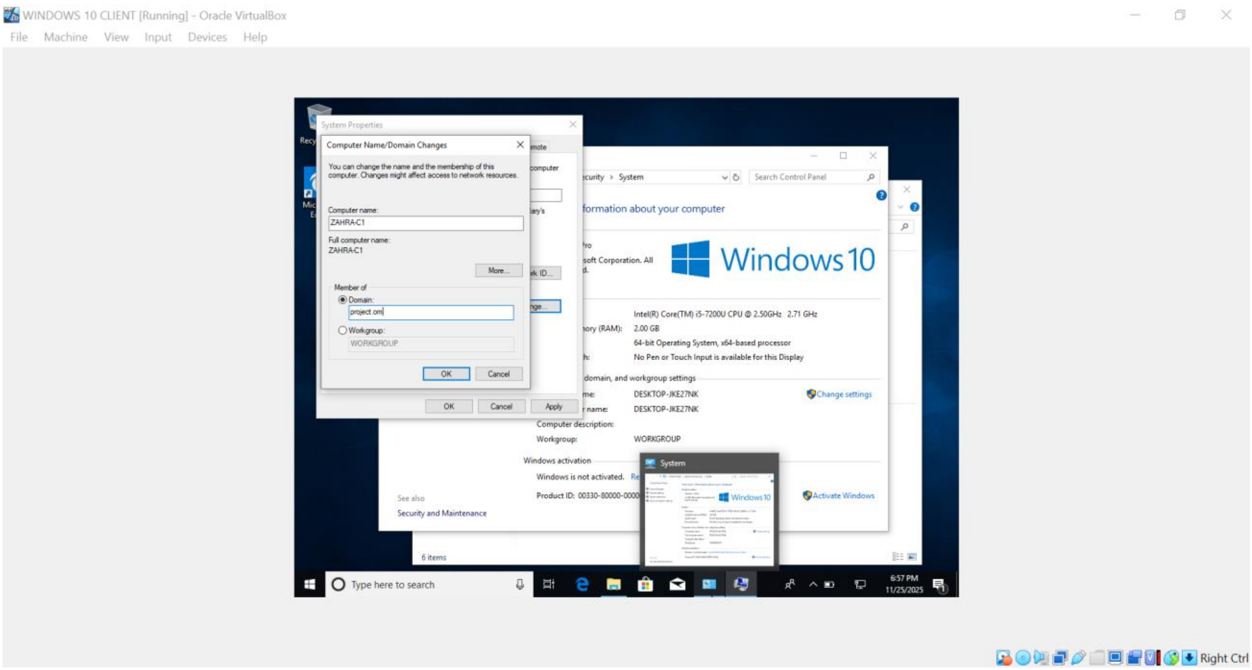

S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
DHCP
File Action View Help
DHCP
zahra-s1.proje
IPv4
Server
Policie
Filters
IPv6
Add a Scope
Actions
IPv4
More Actions
New Scope Wizard
IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.
Configuration settings for DHCP Server
Enter the range of addresses that the scope distributes.
Start IP address: 10 . 0 . 10 . 101
End IP address: 10 . 0 . 10 . 120
Configuration settings that propagate to DHCP Client
Length: 8
Subnet mask: 255 . 0 . 0 . 0
< Back
Next >
Cancel

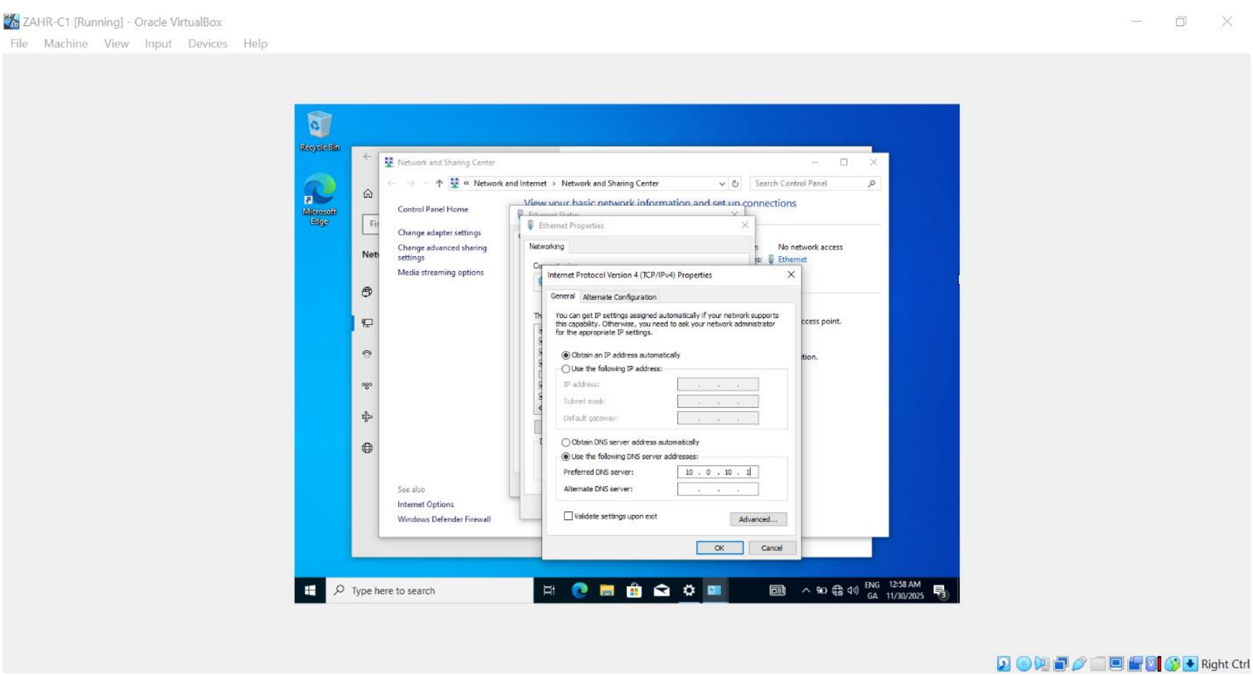

S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
DNS Manager
File Action View Help
DNS
ZAHRA-S1
Forward Lookup Zones
_msdcs.project.om
project.om
projectPrimaryZone
Reverse Lookup Zones
Trust Points
Conditional Forwarders
New Host
Name (uses parent domain name if blank):
X
Fully qualified domain name (FQDN):
X.projectPrimaryZone.
IP address:
192.168.10.2
Create associated pointer (PTR) record
Allow any authenticated user to update DNS records with the same owner name
Add Host
Cancel
Timestam
static
static
Server Manager
ROLES AND SERVER GROUPS
Roles: 4 | Server groups: 1 | Servers total: 1
10:30 PM
11/24/2025
Right Ctrl

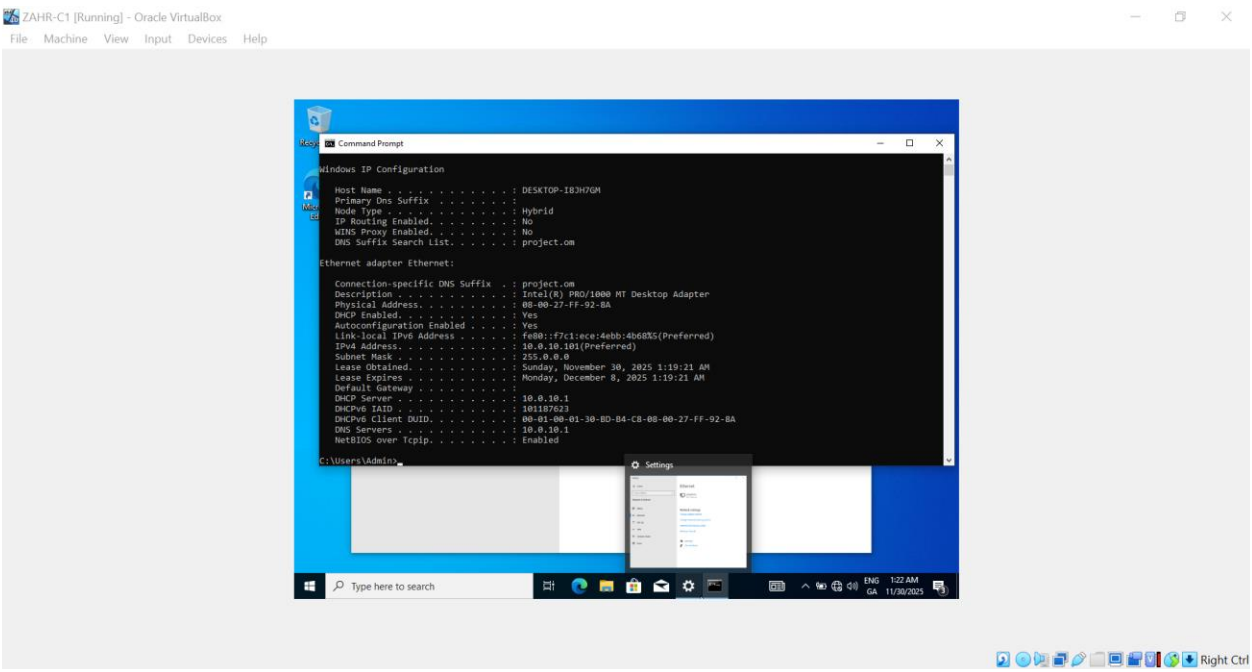

S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
DHCP
File Action View Help
DHCP
zahra-s1.project.om
IPv4
Server Options
Policies
Filters
IPv6
Contents of Scope
Address Pool
Address Leases
Reservations
Scope Options
Policies
Actions
More Actions
10:18 PM
11/24/2025


S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
DNS Manager
File Action View Help
DNS
ZAHRA-S1
Forward Lookup Zones
_msdcs.project.om
project.om
projectPrimaryZone
om
edu
Reverse Lookup Zones
Trust Points
Conditional Forwarders
Name
Type
Data
Timestam
edu
ROLES AND SERVER GROUPS
Roles: 4 | Server groups: 1 | Servers total: 1
10:32 PM
11/24/2025
Right Ctrl

3. **Configure Windows 10 as YourName-C-1, join with the server and get ip through DHCP.**

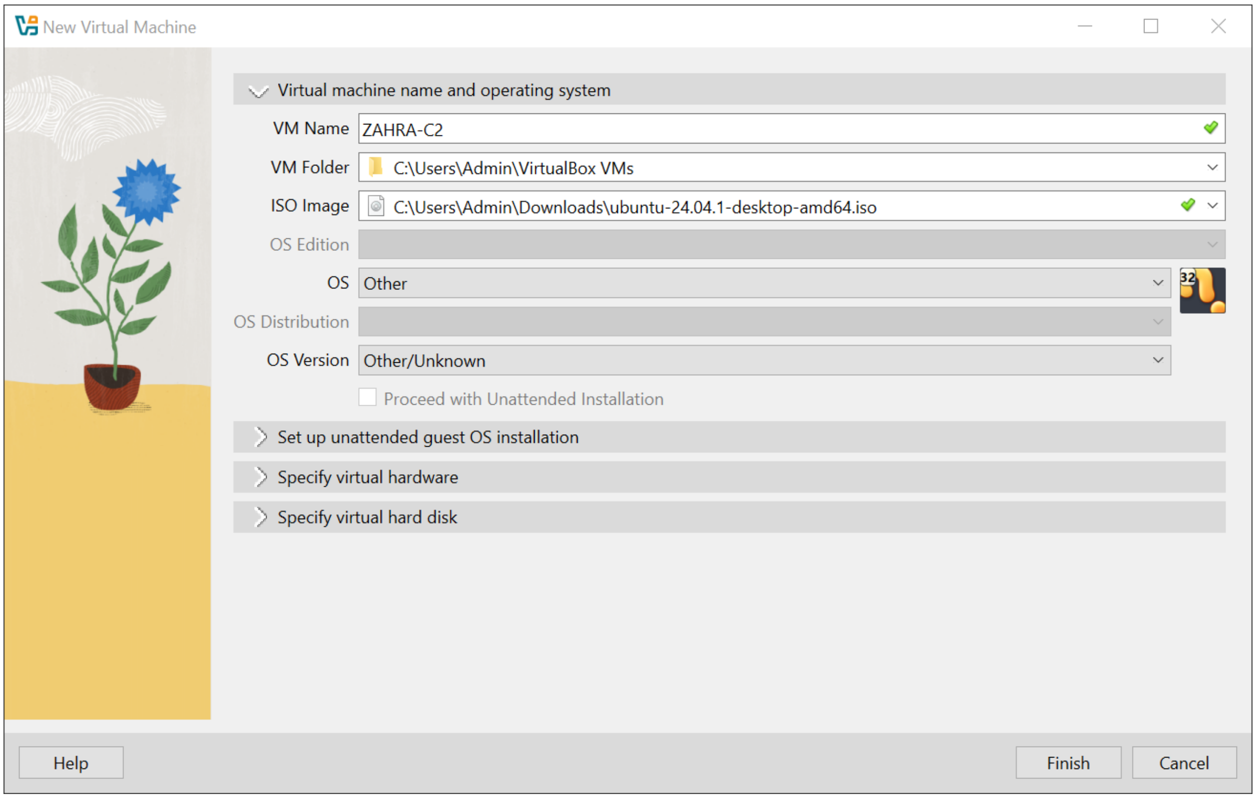


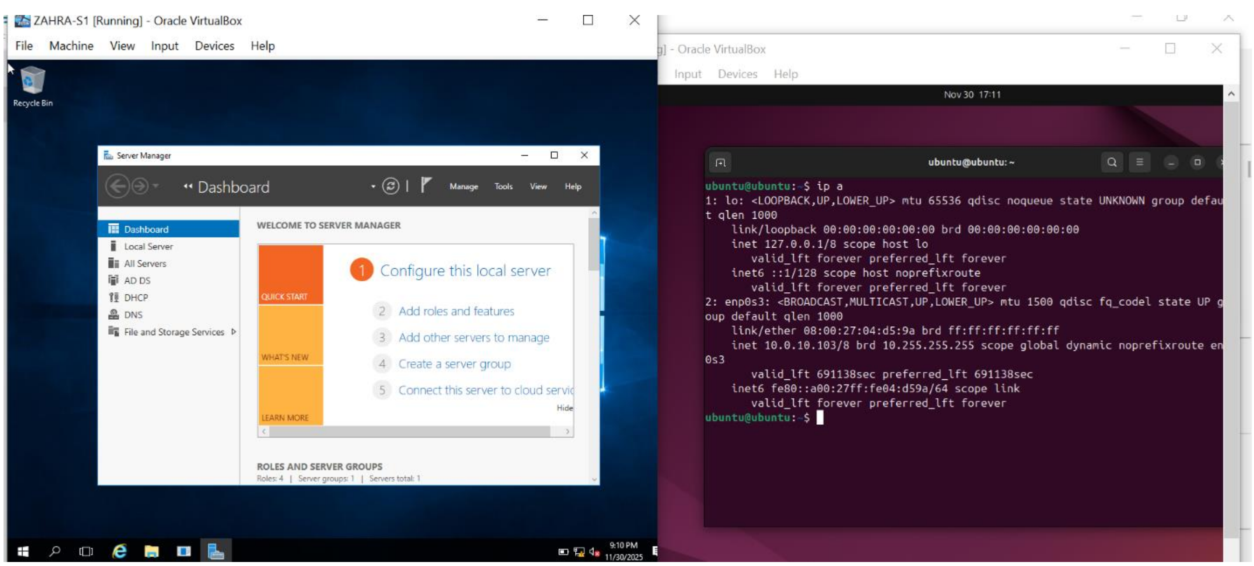




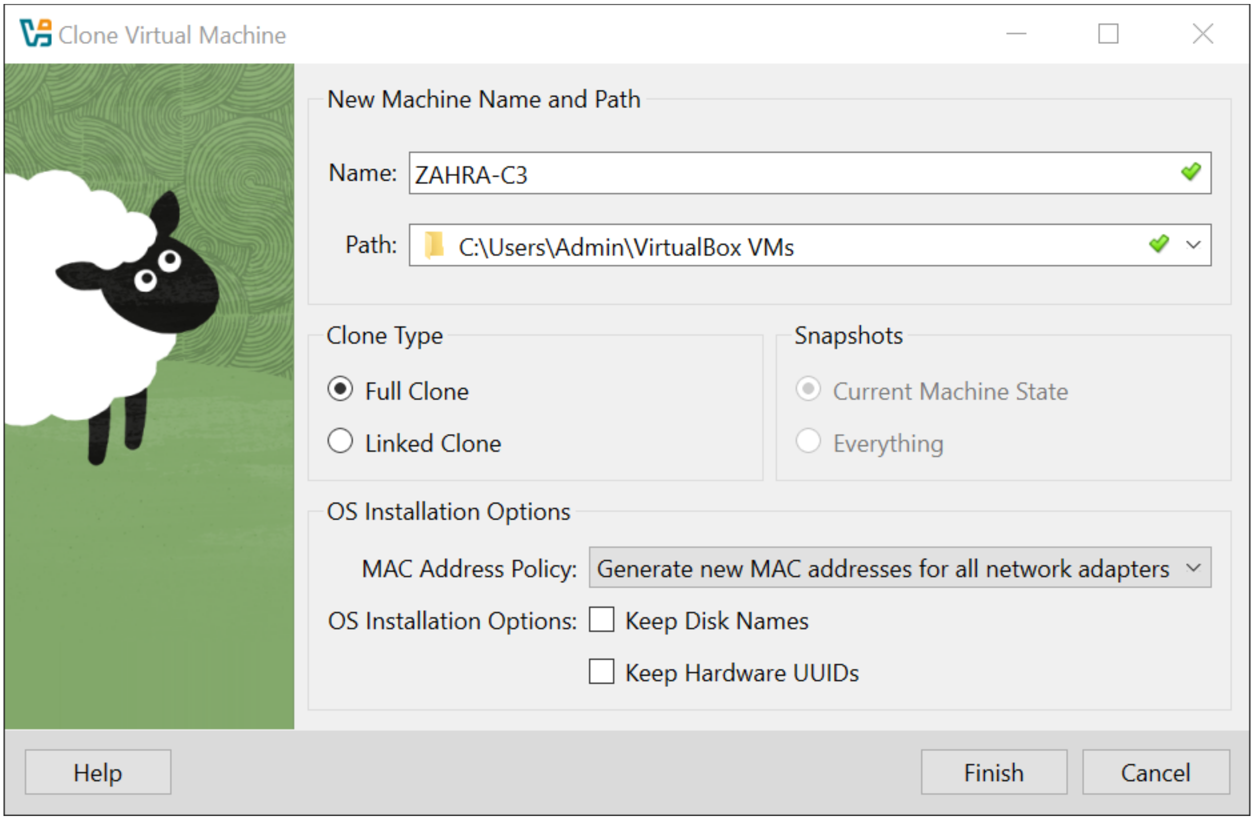
ZAHR-C1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Command Prompt
Windows IP Configuration
Host Name . . . . : DESKTOP-I8JH7GM
Primary Dns Suffix . . . . :
Node Type . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . . : project.om
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : project.om
Description . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-FF-92-8A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::f7c1:ece:4ebb:4b68%5(Preferred)
IPv4 Address. . . . . : 10.0.10.101(Preferred)
Subnet Mask . . . . : 255.0.0.0
Lease Obtained. . . . . : Sunday, November 30, 2025 1:19:21 AM
Lease Expires . . . . : Monday, December 8, 2025 1:19:21 AM
Default Gateway . . . . :
DHCP Server . . . . : 10.0.10.1
DHCPv6 IAID . . . . : 101187623
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-BD-B4-C8-08-00-27-FF-92-8A
DNS Servers . . . . : 10.0.10.1
NetBIOS over Tcpip. . . . . : Enabled
C:\Users\Admin>
Settings
Type here to search
ENG
GA
1:22 AM
11/30/2025
Right Ctrl

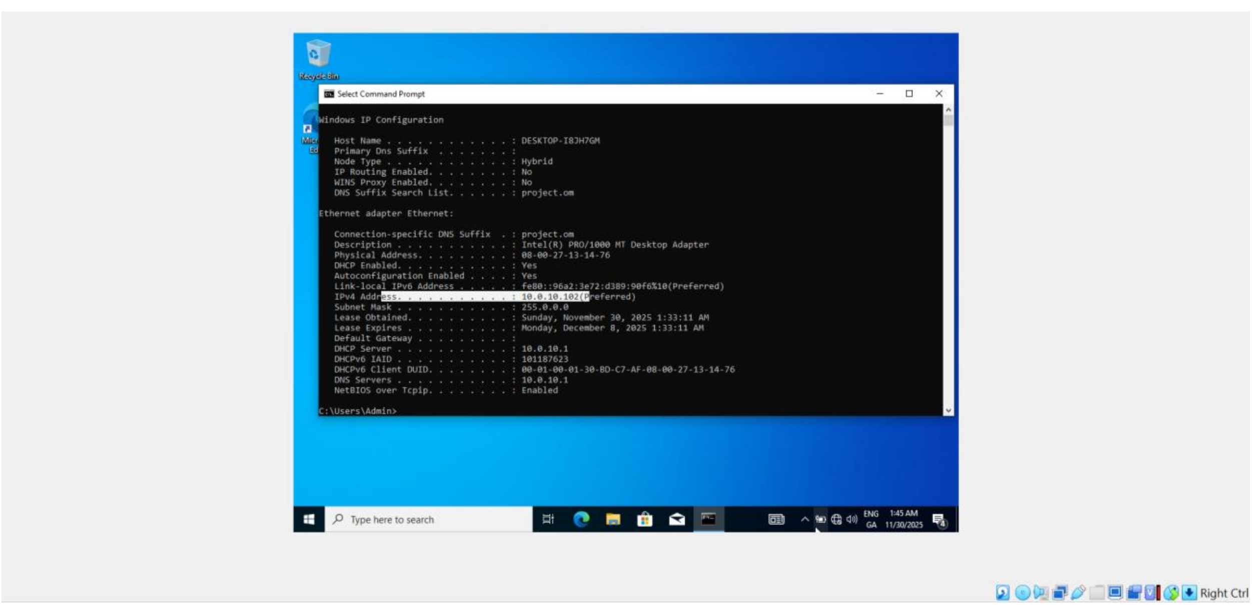
4. Install and configure Ubuntu asYourName-C-2.



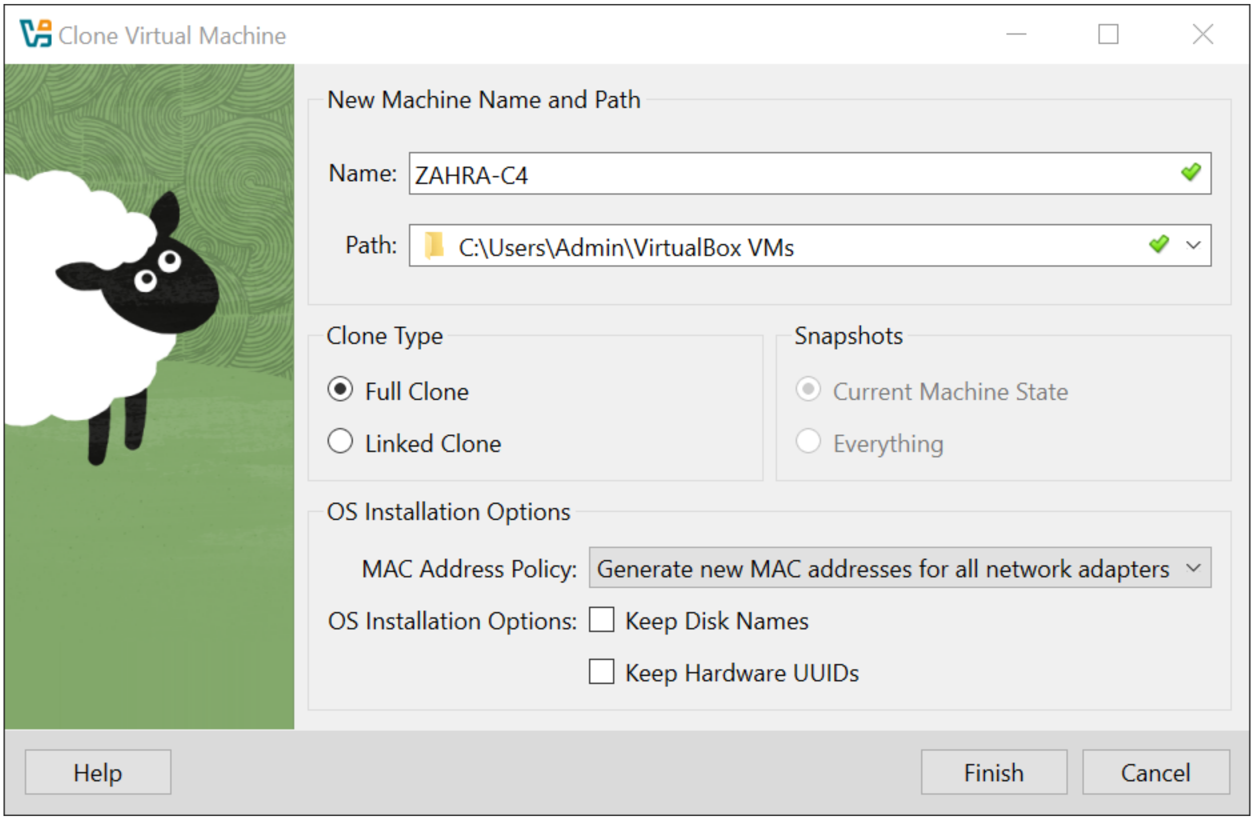


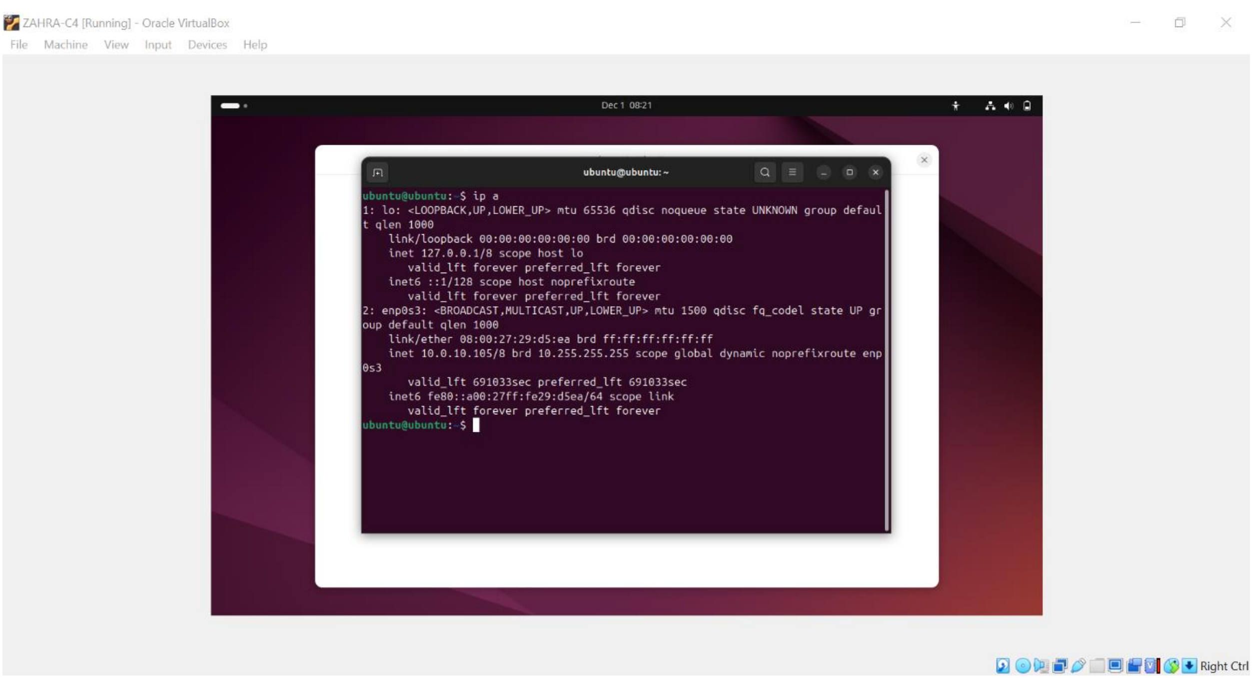
5. Make a clone of Windows 10 as YourName-C-3, should get IP address from DHCP server.





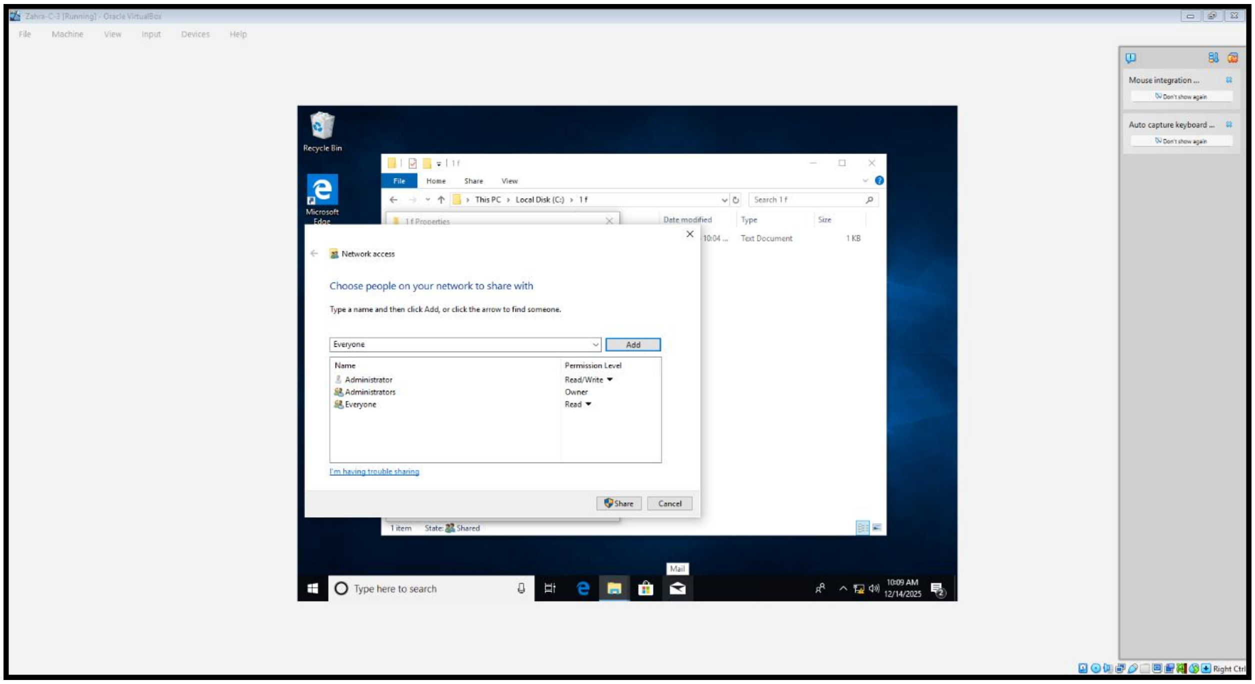
6. Make a clone of Ubuntu as YourName-C-4 should get IP address from DHCP server.



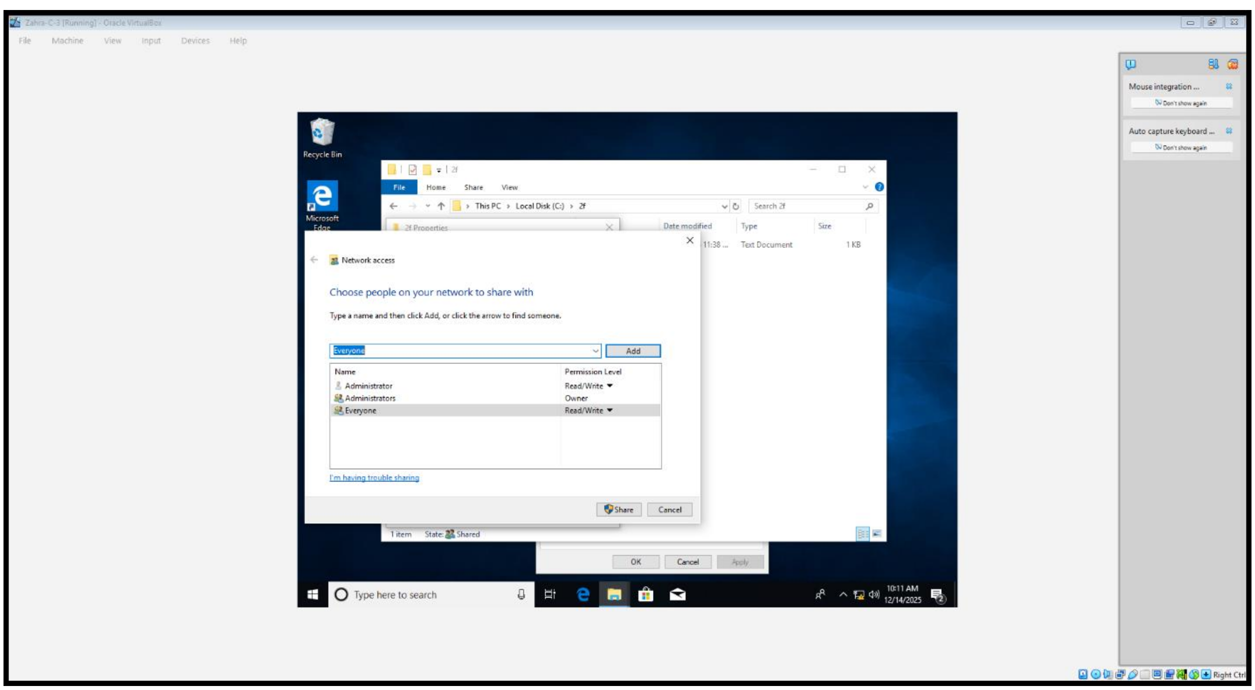


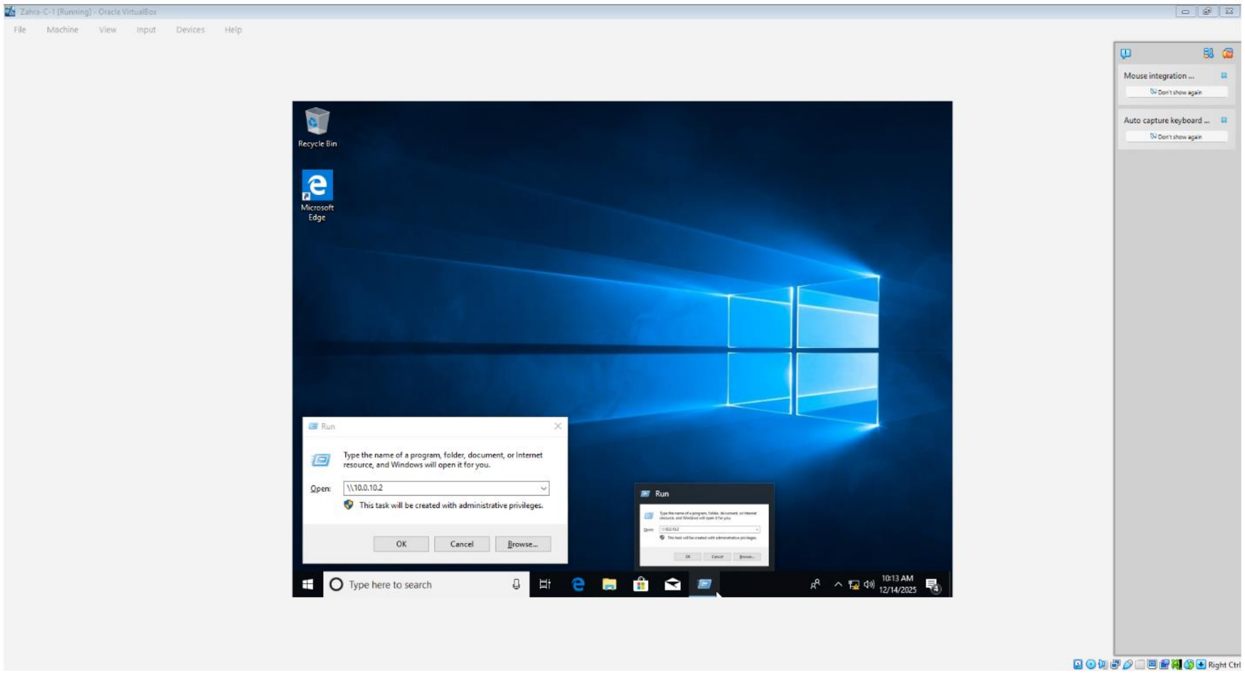
7.Create 2-2 folders in all windows clients, share all created folders. User should be able to copy data in shared folder from any Windows Client.

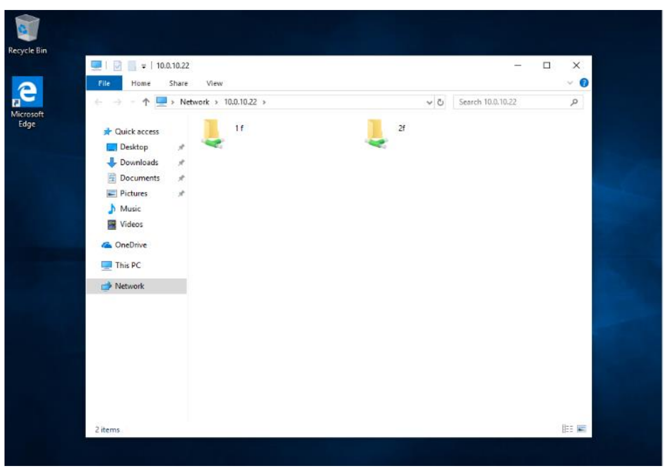
Folder(1f)-Read



Folder(2f)-Read/write:

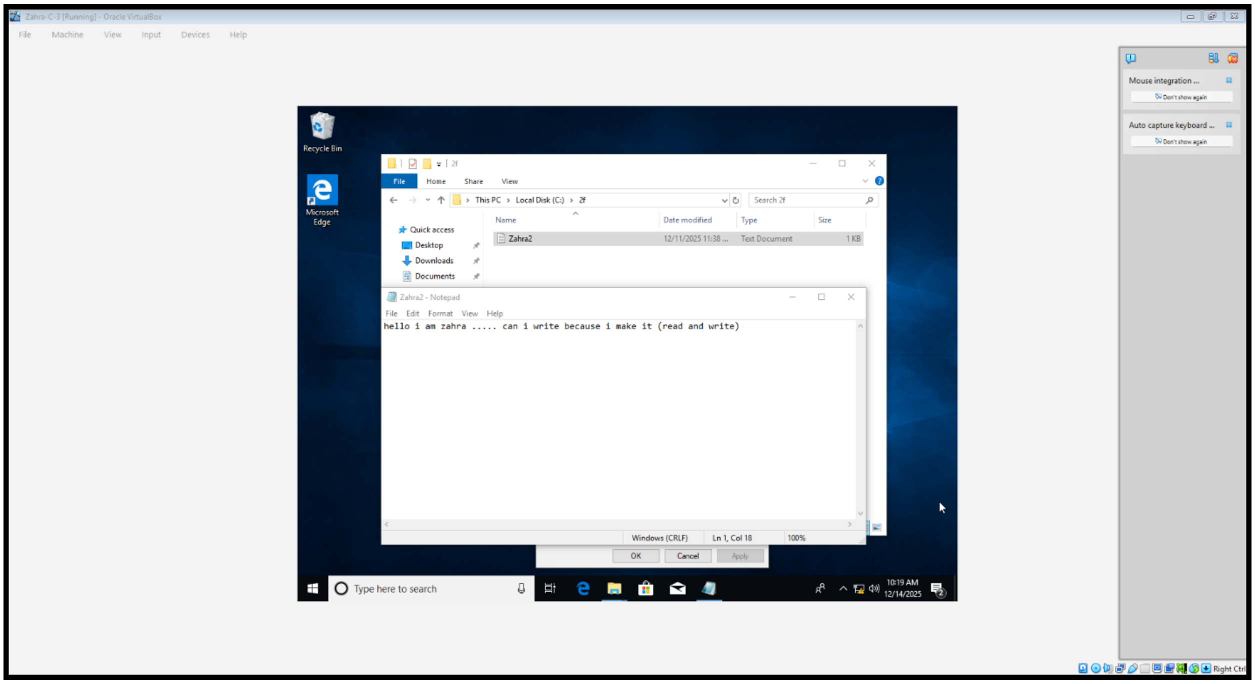



Zahra-C-1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mouse integration ...
Auto capture keyboard ...
Recycle Bin
Microsoft Edge
Run
Type the name of a program, folder, document, or Internet resource, and Windows will open it for you.
Open: \\10.0.10.2
This task will be created with administrative privileges.
OK Cancel Browse...
Type here to search
10:13 AM
12/14/2025
Right Ctrl

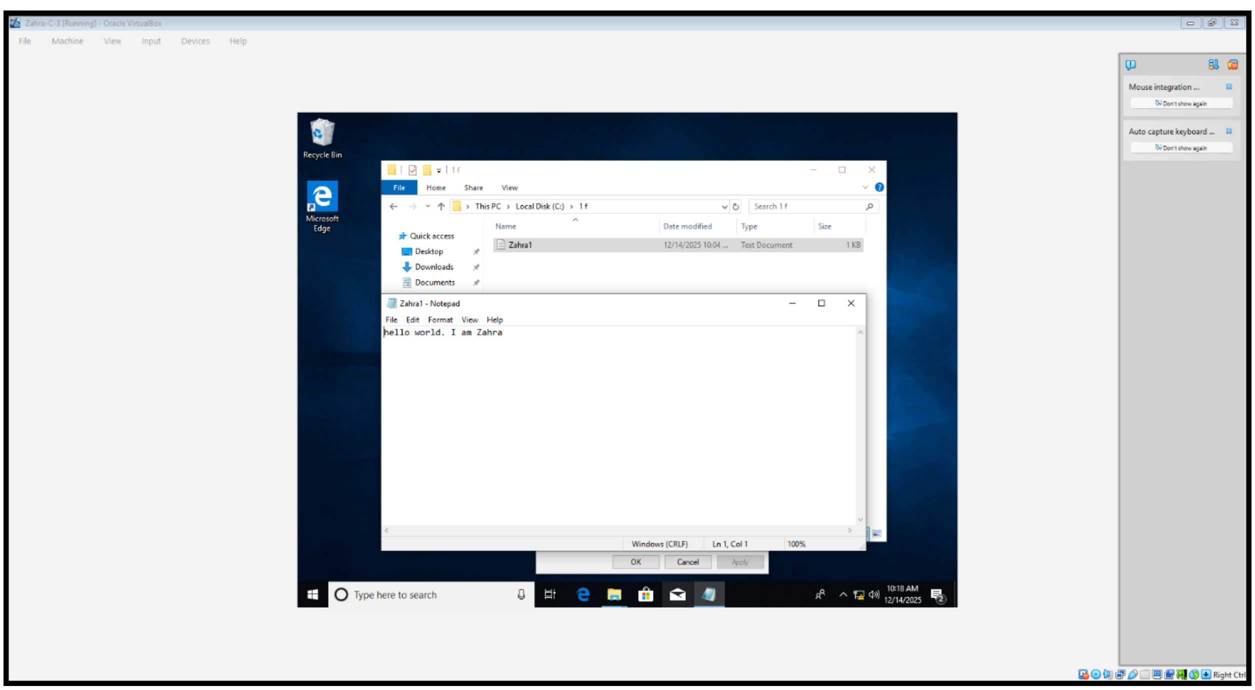

Recycle Bin
Microsoft Edge
10.0.10.22
File Home Share View
Network › 10.0.10.22 ›
Search 10.0.10.22
Quick access
Desktop
Downloads
Documents
Pictures
Music
Videos
OneDrive
This PC
Network
1f
2f
2 items

8. Disable write permission, try to do changes in folders and show result.

--we can copy and modification:

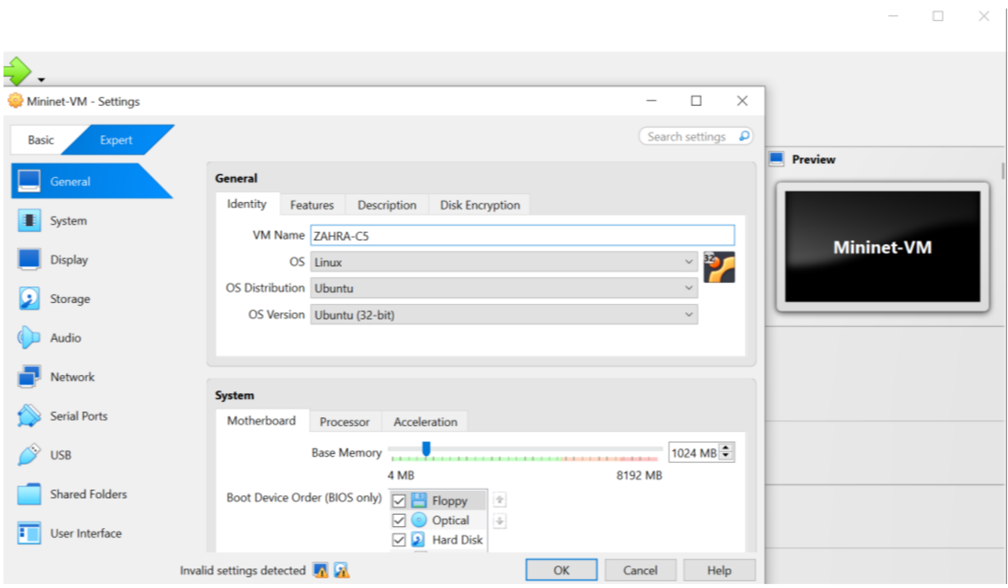


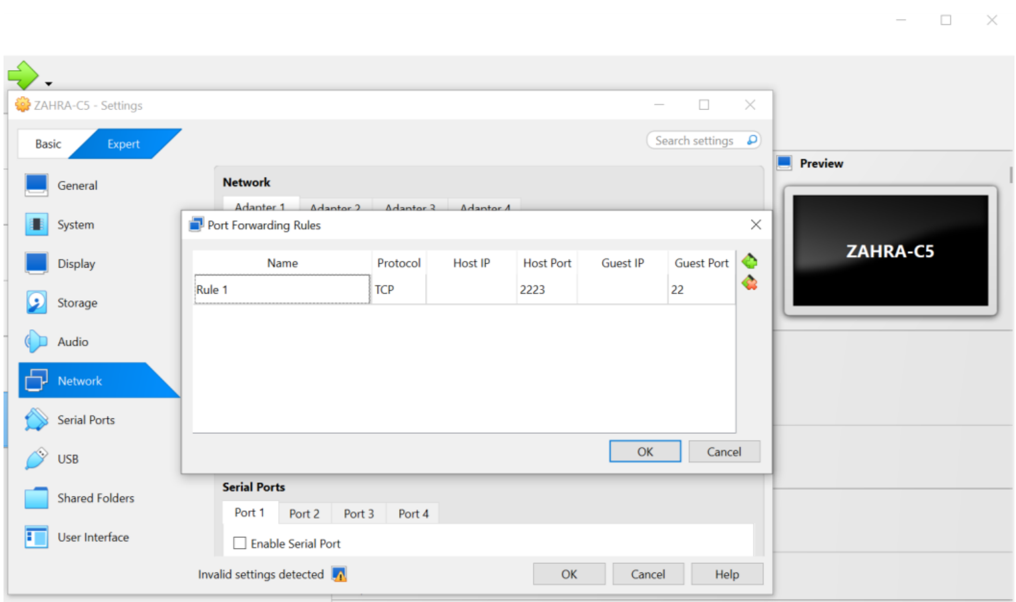
--we cant copy any think only read:



9.Download and Install Mininet, name it as YourName-C-5.

----NO FILE IOS IN ELERNING:





10.Configure the topology with some Python commands in mininet.

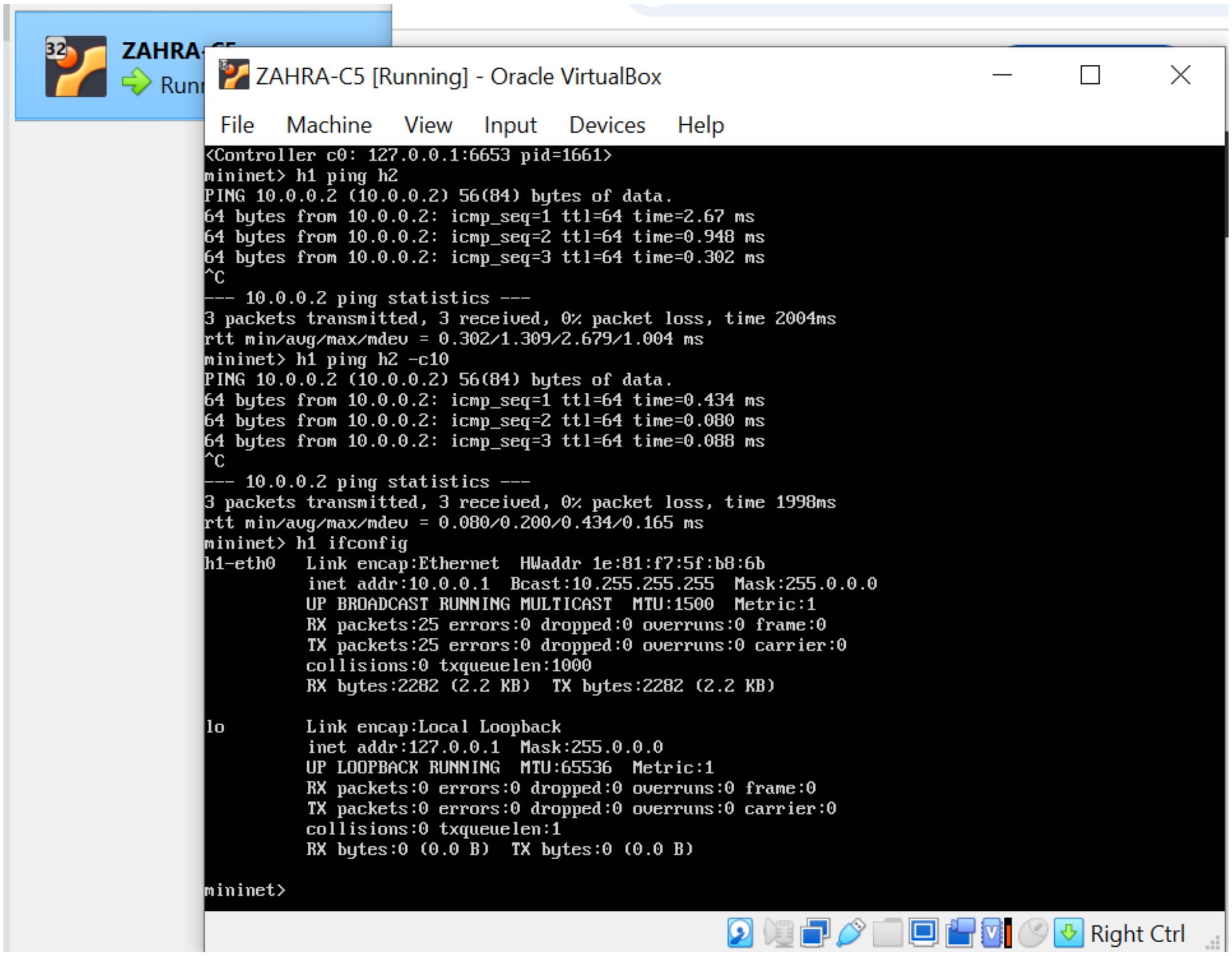
Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ssh mininet@localhost -p 2223
The authenticity of host '[localhost]:2223 ([127.0.0.1]:2223)' can't be established.
ED25519 key fingerprint is SHA256:5BTWlpXOVa8JMzvTg3xMTJkkbU/dvuIf+lrKxkSsY3E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2223' (ED25519) to the list of known hosts.
mininet@localhost's password:
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-142-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Dec  9 00:28:53 2025
mininet@mininet-vm:~$ ls
mininet  oflops  oftest  openflow  pox
mininet@mininet-vm:~$ cd mininet
mininet@mininet-vm:~/mininet$ ls
bin           custom  doc       INSTALL  Makefile  mn.1    mnexec.1  README.md  util
CONTRIBUTORS  debian  examples  LICENSE  mininet   mnexec  mnexec.c  setup.py
mininet@mininet-vm:~/mininet$ exit
logout
Connection to localhost closed.

C:\Users\Admin>
```

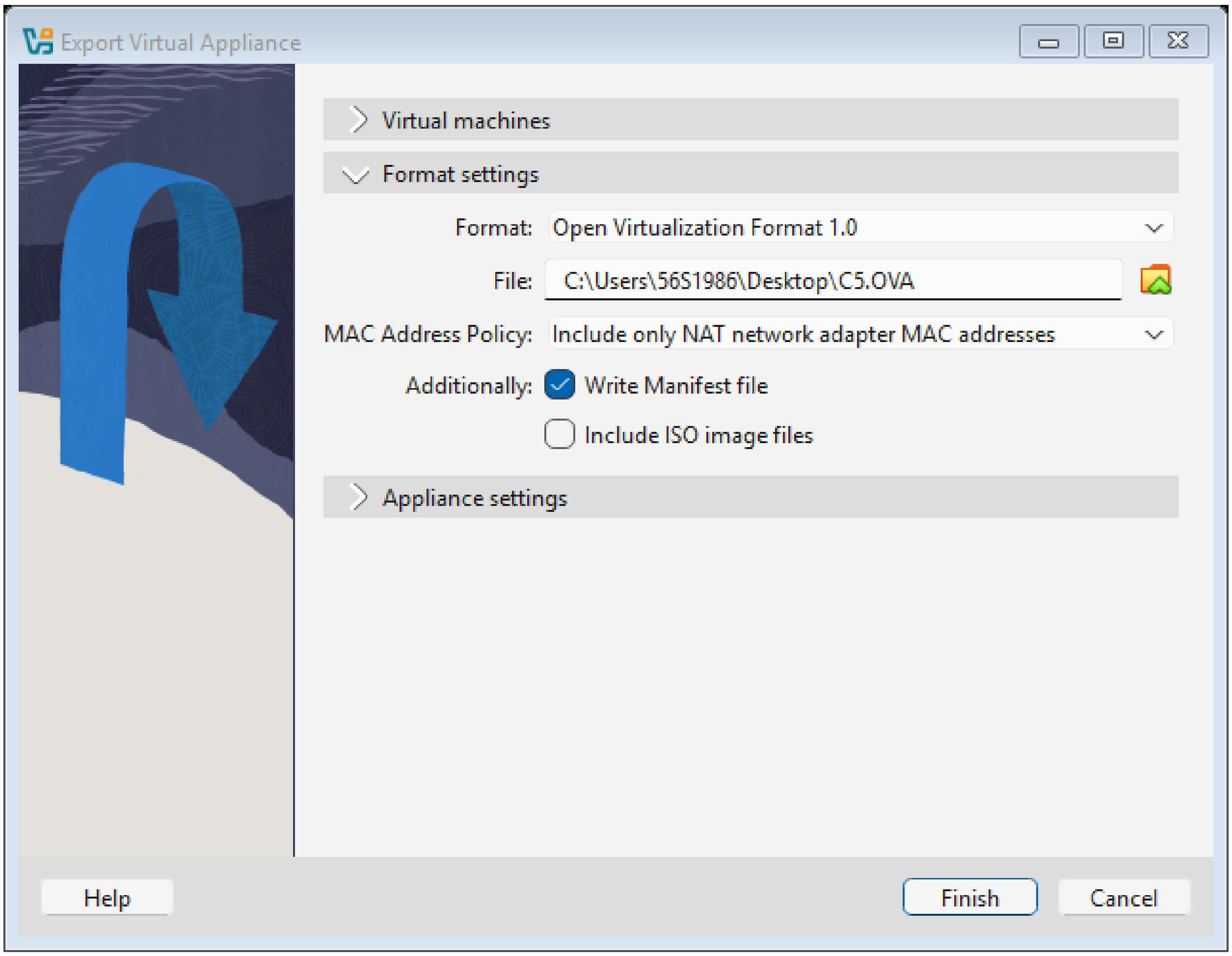
```
mininet> nodes
available nodes are:
c0 h1 h2 s1
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1668>
<Host h2: h2-eth0:10.0.0.2 pid=1671>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=1677>
<Controller c0: 127.0.0.1:6653 pid=1661>
mininet> h
```

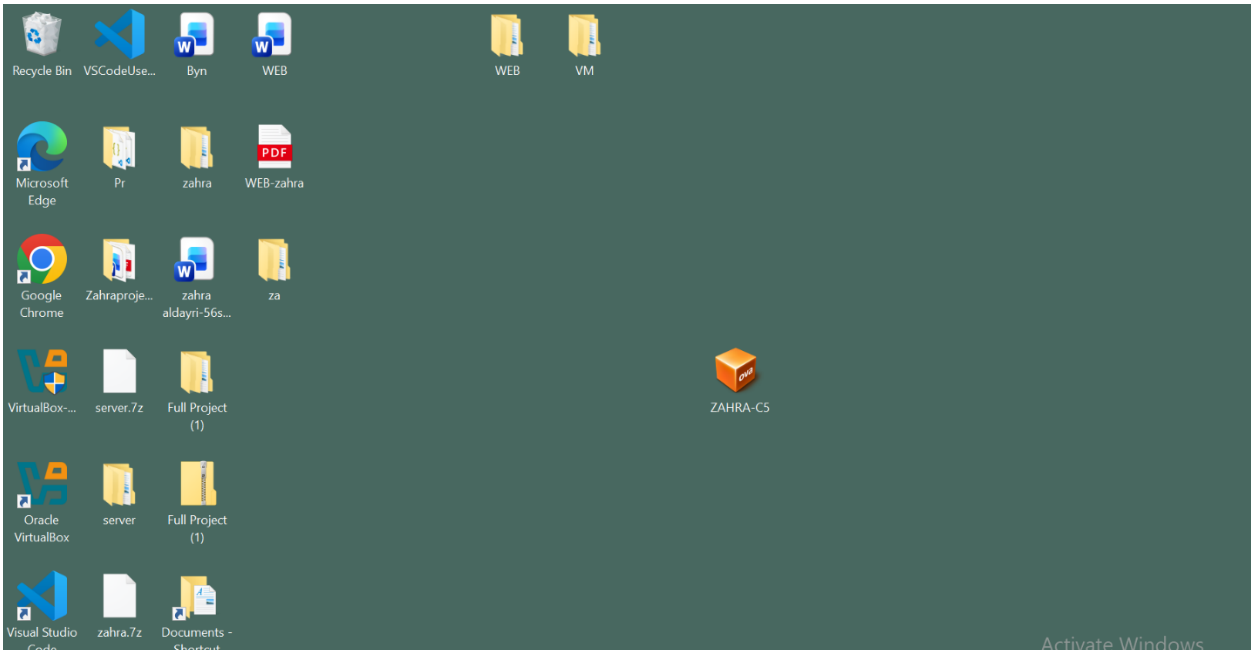
Right Ctrl

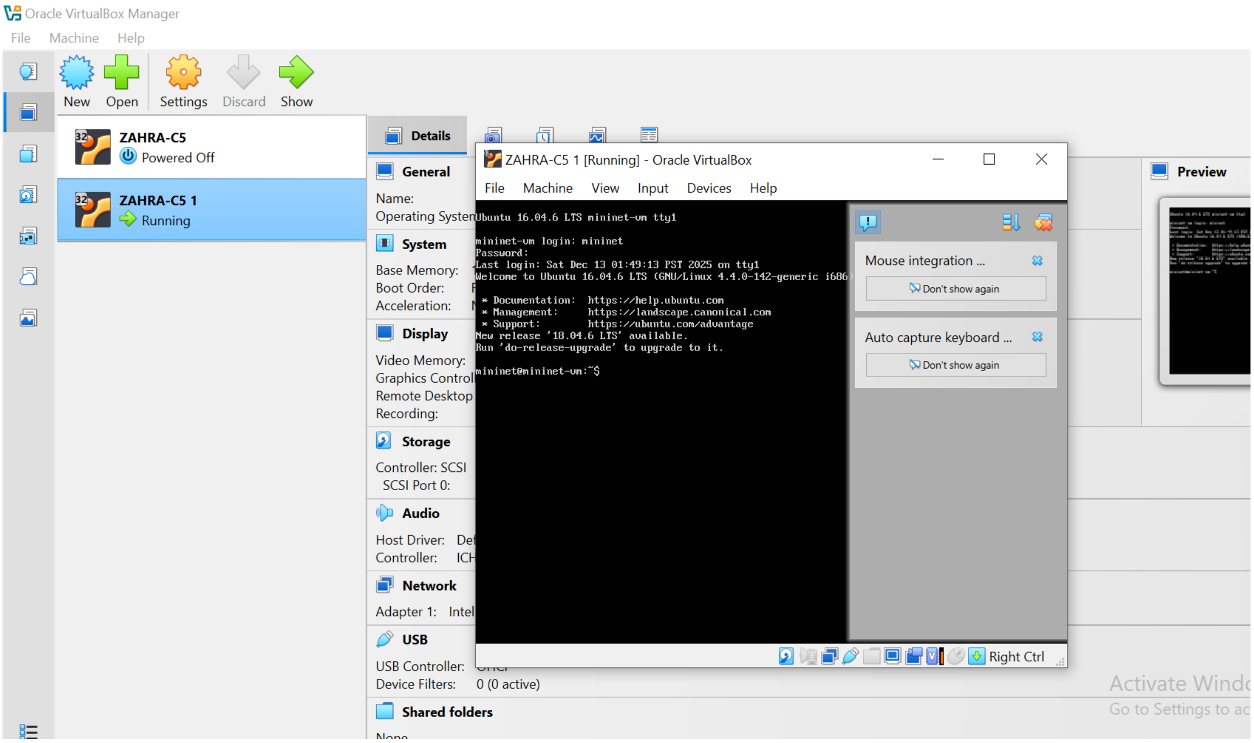
11. Show communication between H1, H2 and CO. (By using ping).



12.Create an OVF of mininet, then delete the installed VM, and then import created OVF and show its working running mode.

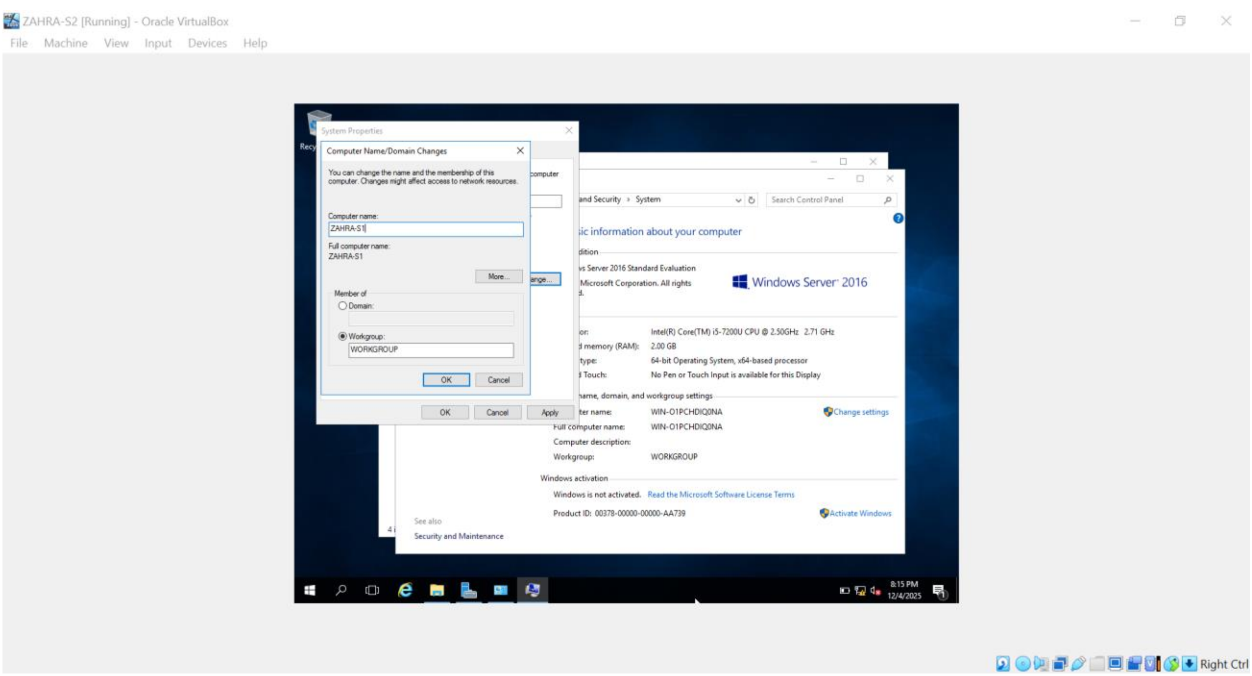


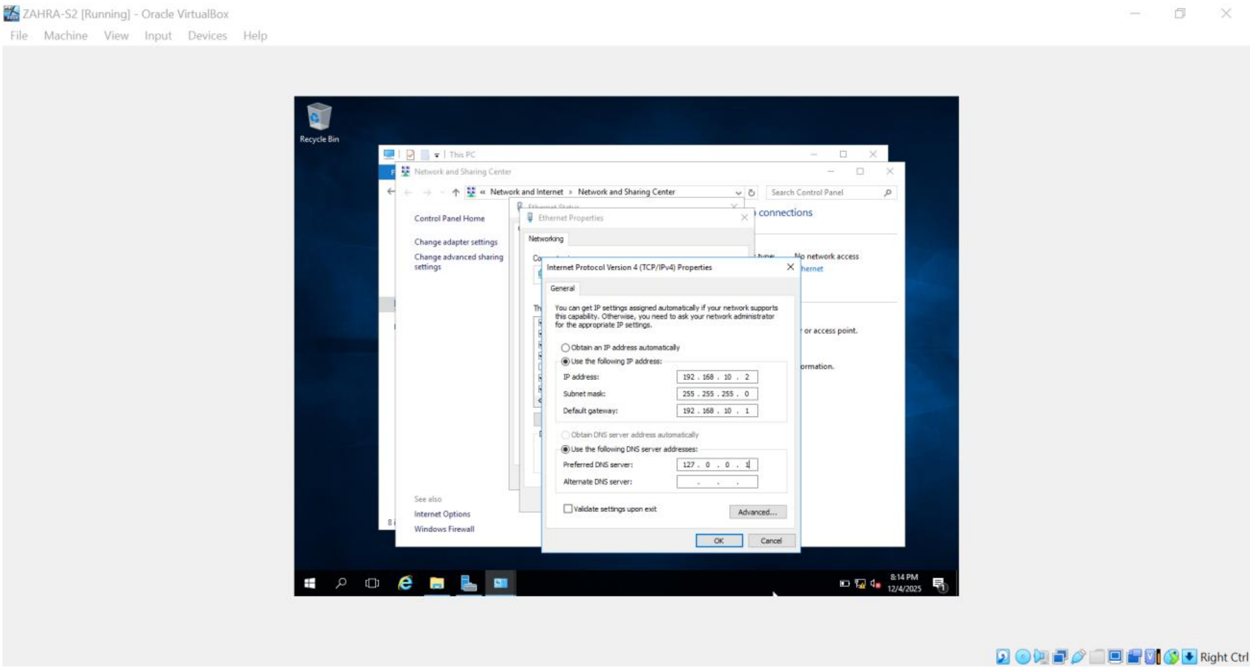

Recycle Bin
VSCodeUse...
Byn
WEB
WEB
VM
Microsoft Edge
Pr
zahra
WEB-zahra
Google Chrome
Zahraproje...
zahra aldayri-56s...
za
VirtualBox-...
server.7z
Full Project (1)
ZAHRA-CS
Oracle VirtualBox
server
Full Project (1)
Visual Studio
zahra.7z
Documents -



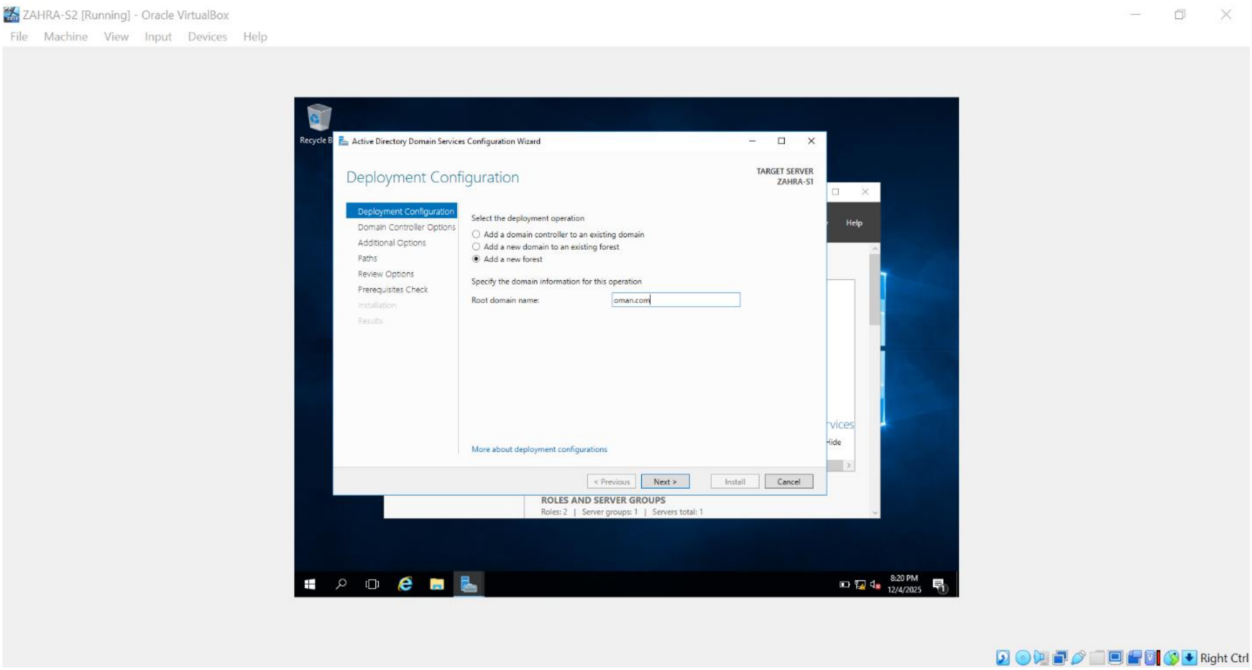
13. Import Windows Server 2016 by the name YourName-S-2.

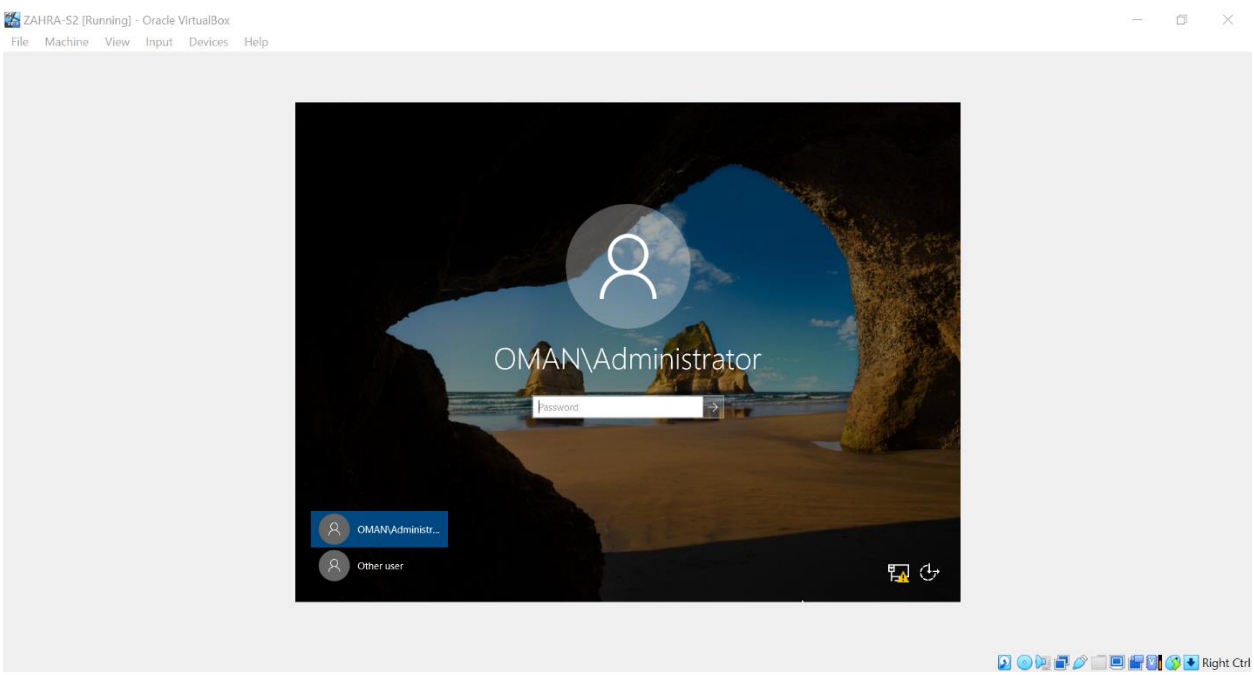
14. For ID conflict do necessary SystemPrep procedure.

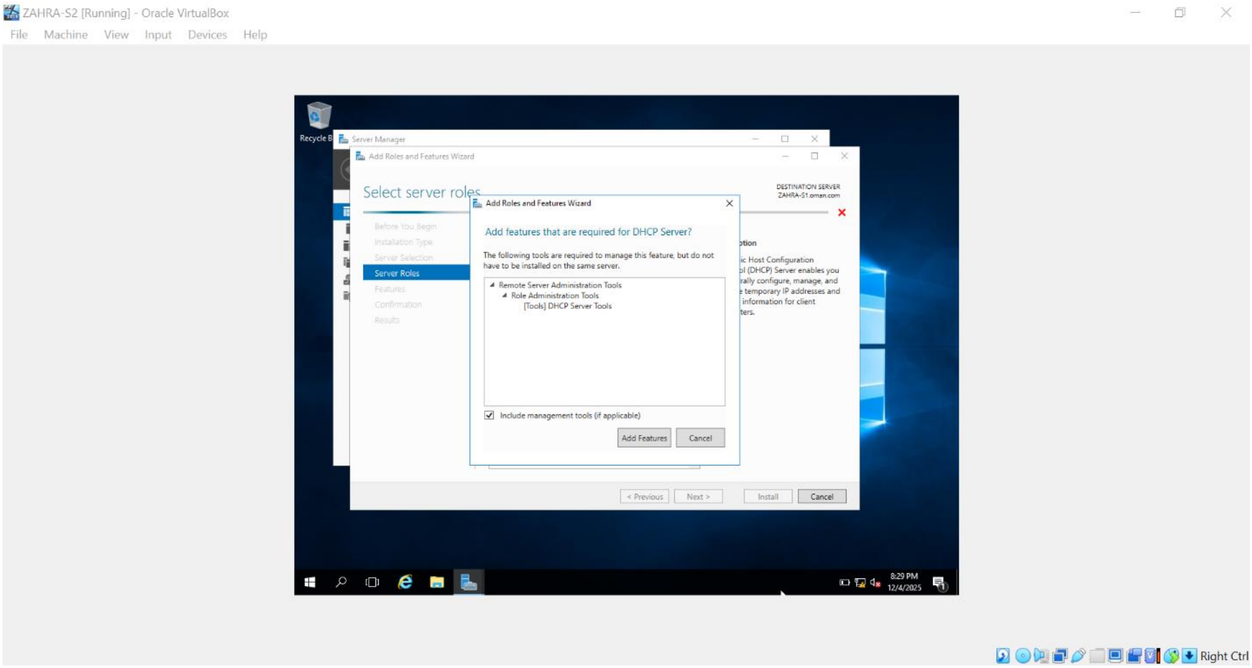


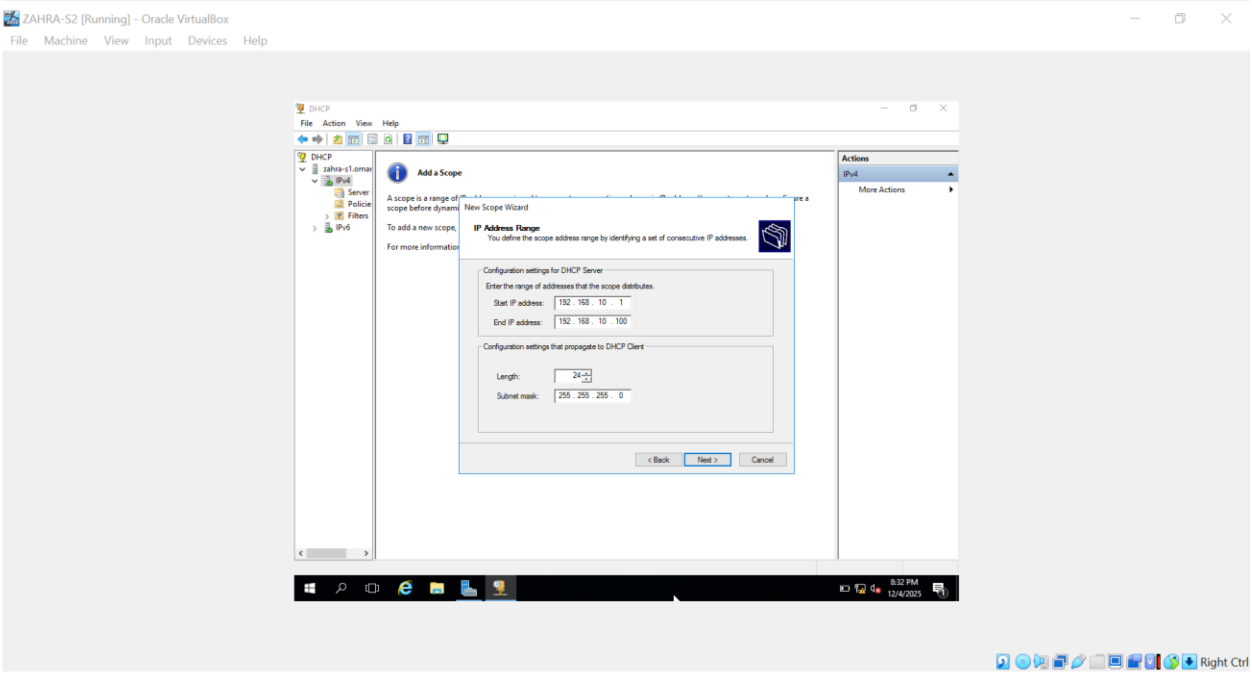

ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Network and Sharing Center
Network and Internet > Network and Sharing Center
Search Control Panel
Control Panel Home
Change adapter settings
Change advanced sharing settings
Ethernet Properties
Networking
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 192 . 168 . 10 . 2
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 192 . 168 . 10 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server: 127 . 0 . 0 . 1
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
No network access
See also
Internet Options
Windows Firewall
8:14 PM
12/4/2025
Right Ctrl

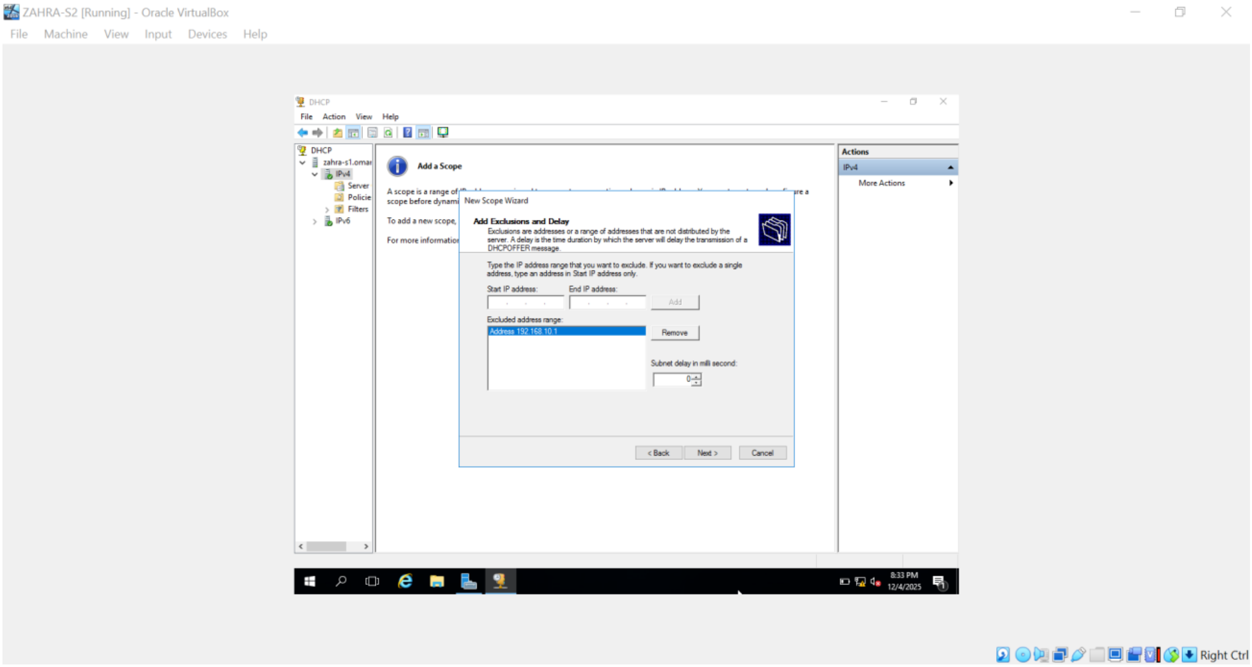
15.Configure YourName-S-2 ADDC, DNS (oman.com), DHCP and connect YourName-S-1, both client should receive IP address automatically.

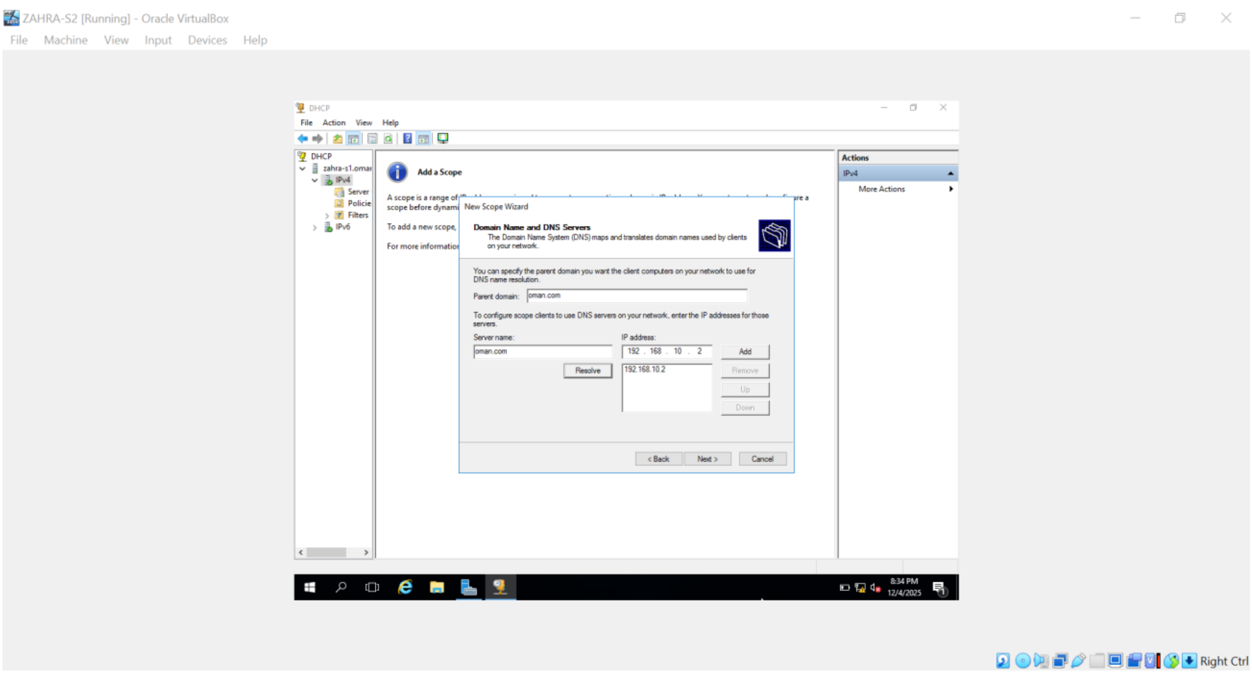


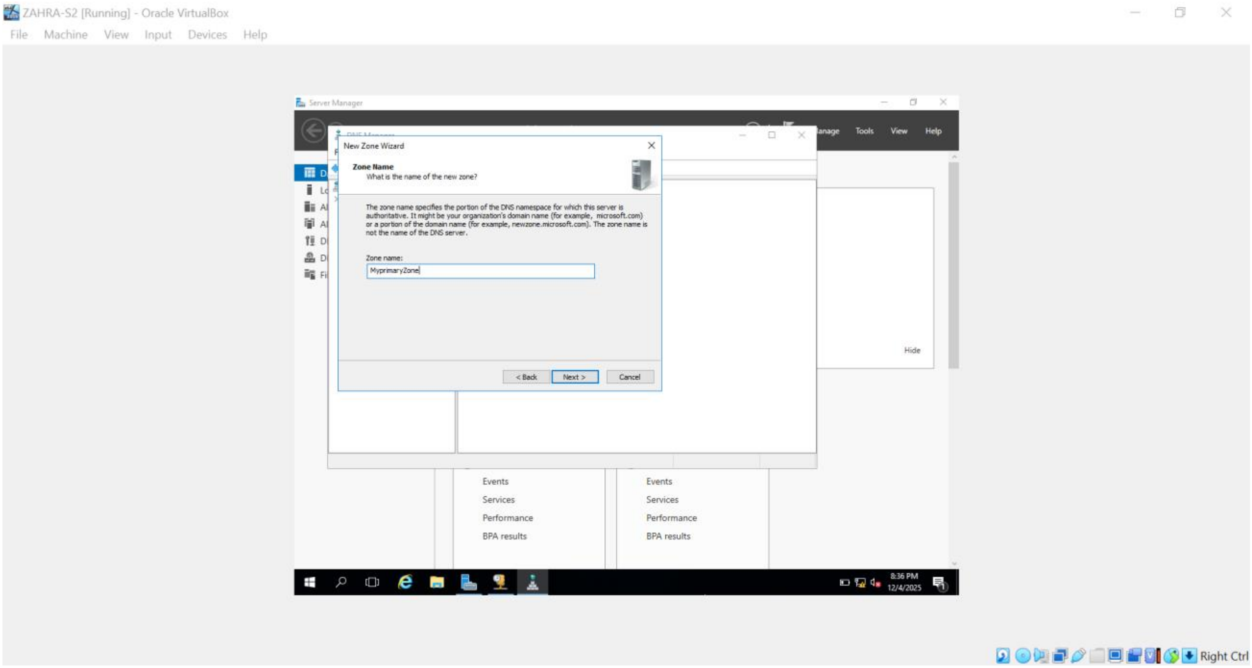



ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Server Manager
Add Roles and Features Wizard
Select server roles
DESTINATION SERVER
ZAHRA-S1.oman.com
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Add Roles and Features Wizard
Add features that are required for DHCP Server?
The following tools are required to manage this feature, but do not have to be installed on the same server.
Remote Server Administration Tools
Role Administration Tools
[Tools] DHCP Server Tools
Include management tools (if applicable)
Add Features
Cancel
< Previous
Next >
Install
Cancel
8:29 PM
12/4/2025
Right Ctrl

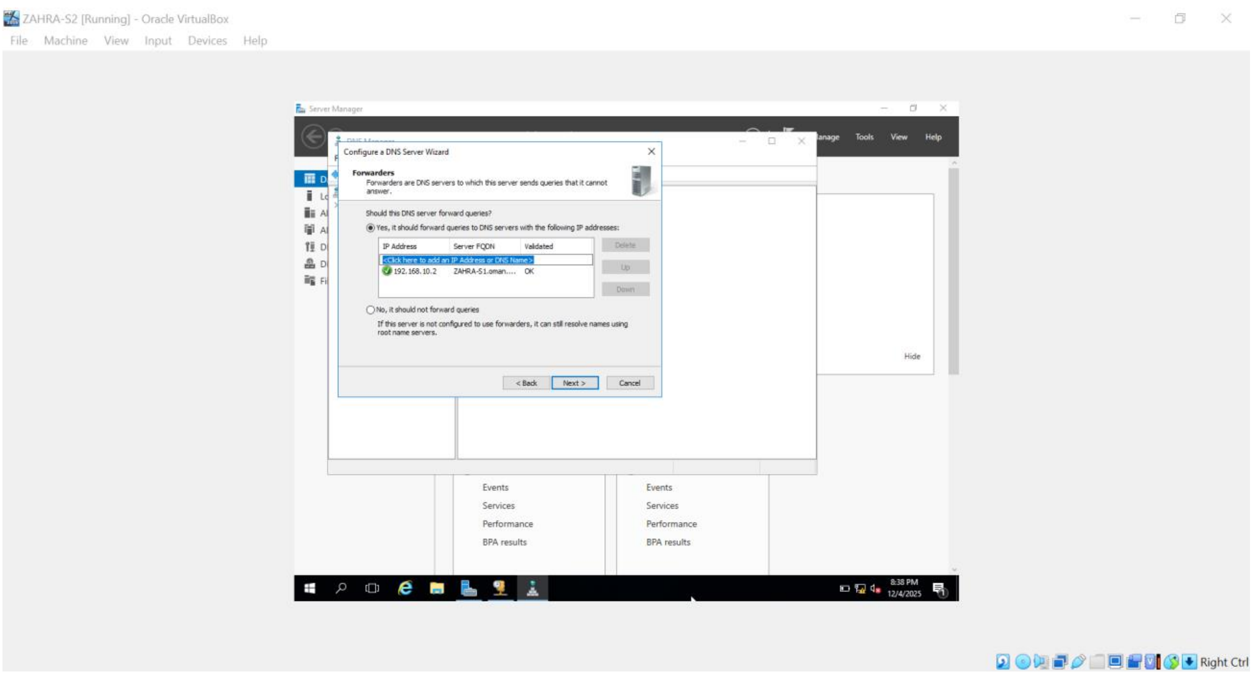

ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
DHCP
File Action View Help
Add a Scope
New Scope Wizard
IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.
Configuration settings for DHCP Server
Enter the range of addresses that the scope distributes.
Start IP address: 192 . 168 . 10 . 1
End IP address: 192 . 168 . 10 . 100
Configuration settings that propagate to DHCP Client
Length: 24
Subnet mask: 255 . 255 . 255 . 0
< Back
Next >
Cancel
Actions
IPv4
More Actions
8:32 PM
12/4/2025
Right Ctrl


ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
DHCP
File Action View Help
Add a Scope
New Scope Wizard
Add Exclusions and Delay
Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCPOFFER message.
Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.
Start IP address:
End IP address:
Add
Excluded address range:
Address 192.168.10.1
Remove
Subnet delay in milli second:
< Back
Next >
Cancel
Actions
IPv4
More Actions
Right Ctrl

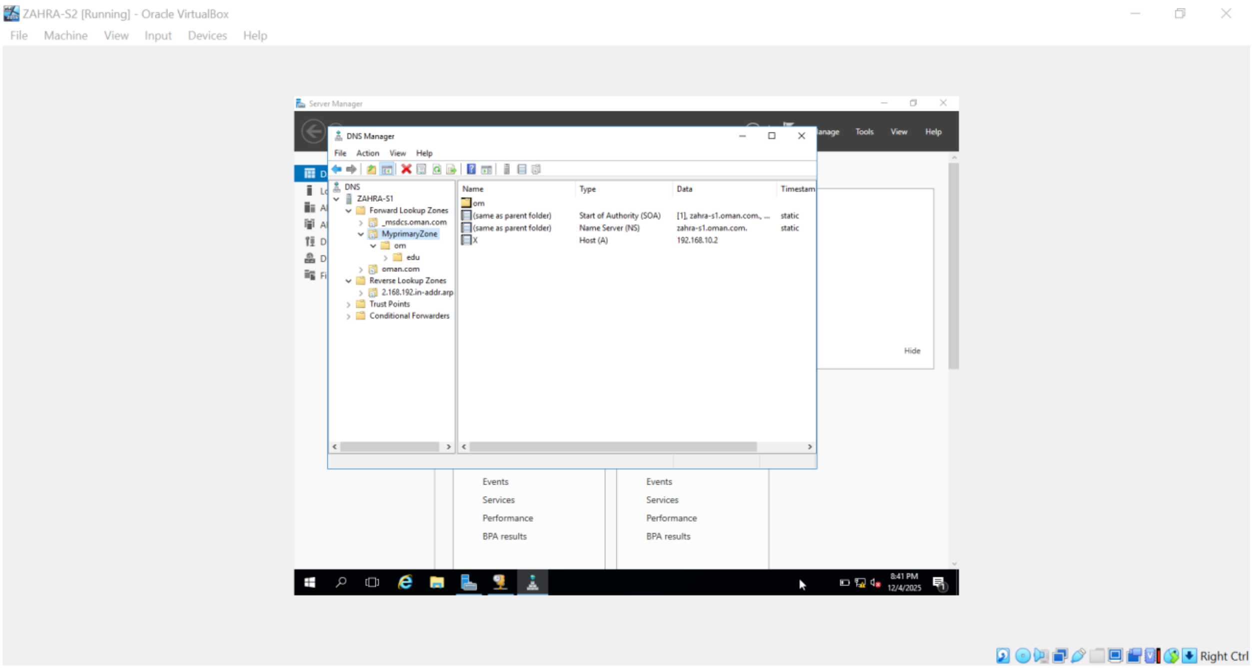

ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
DHCP
File Action View Help
Add a Scope
New Scope Wizard
Domain Name and DNS Servers
The Domain Name System (DNS) maps and translates domain names used by clients on your network.
You can specify the parent domain you want the client computers on your network to use for DNS name resolution.
Parent domain: oman.com
To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.
Server name:
IP address:
oman.com
192 . 168 . 10 . 2
Add
Resolve
192.168.10.2
Remove
Up
Down
< Back
Next >
Cancel
Actions
IPv4
More Actions
Right Ctrl



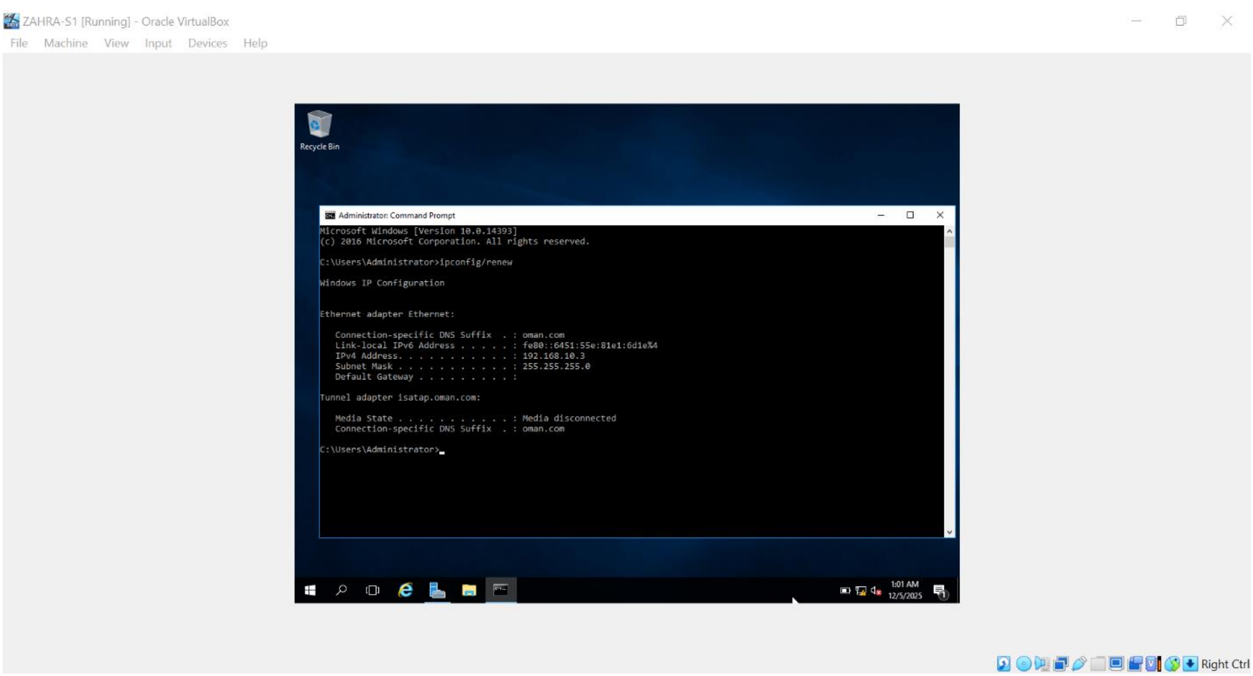
ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
New Zone Wizard
Zone Name
What is the name of the new zone?
The zone name specifies the portion of the DNS namespace for which this server is authoritative. It might be your organization's domain name (for example, microsoft.com) or a portion of the domain name (for example, newzone.microsoft.com). The zone name is not the name of the DNS server.
Zone name:
MyprimaryZone
< Back
Next >
Cancel
Events
Services
Performance
BPA results
8:36 PM
12/4/2025
Right Ctrl

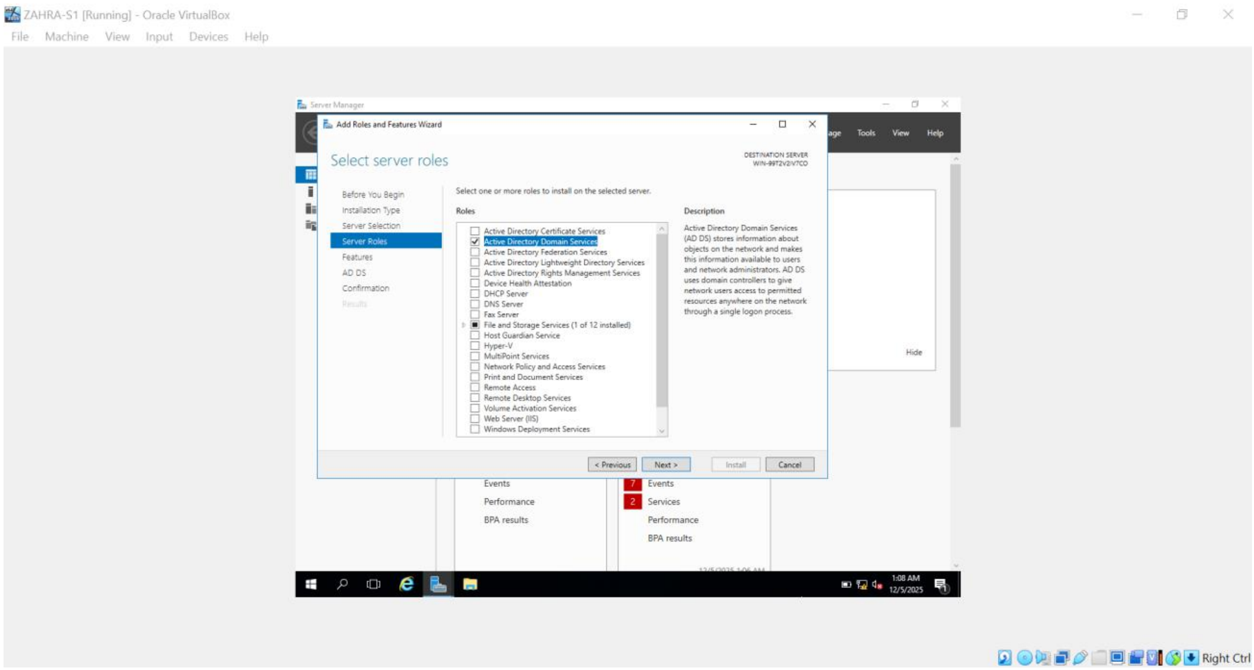


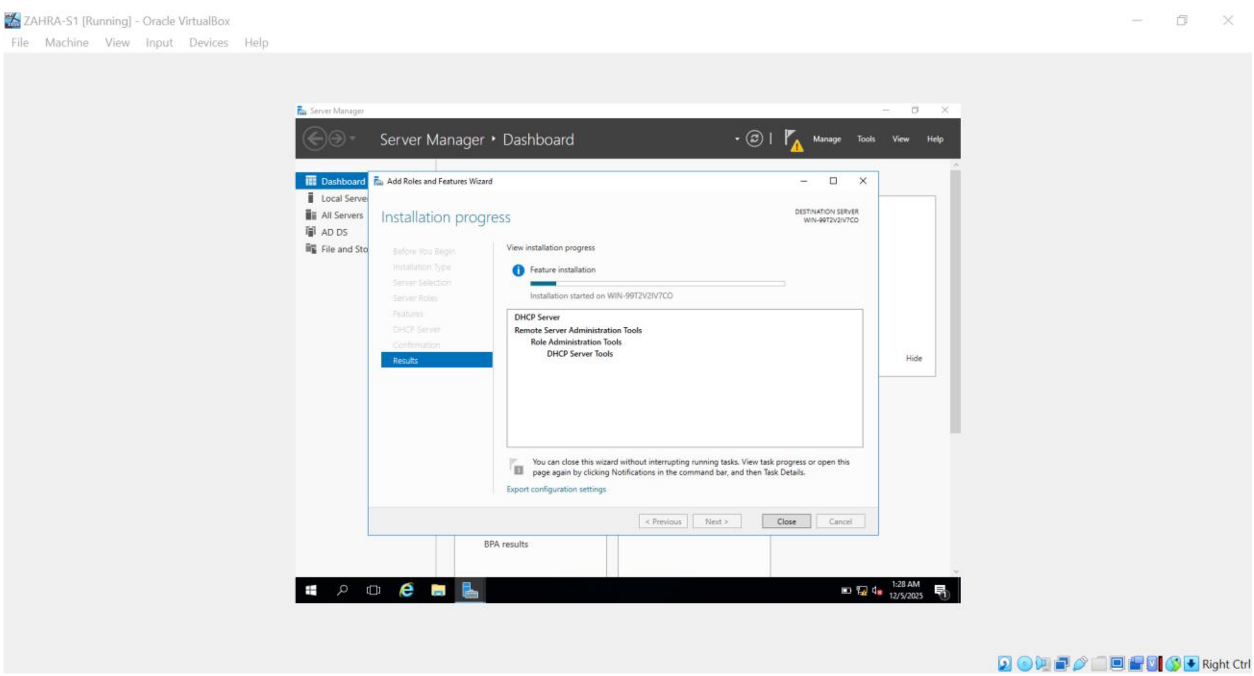
ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Configure a DNS Server Wizard
Forwarders
Forwarders are DNS servers to which this server sends queries that it cannot answer.
Should this DNS server forward queries?
Yes, it should forward queries to DNS servers with the following IP addresses:
IP Address Server FQDN Validated
<Click here to add an IP Address or DNS Name>
192.168.10.2 ZAHRA-S1.oman.... OK
Delete
Up
Down
No, it should not forward queries
If this server is not configured to use forwarders, it can still resolve names using root name servers.
< Back
Next >
Cancel
Events
Services
Performance
BPA results
8:38 PM
12/4/2025
Right Ctrl

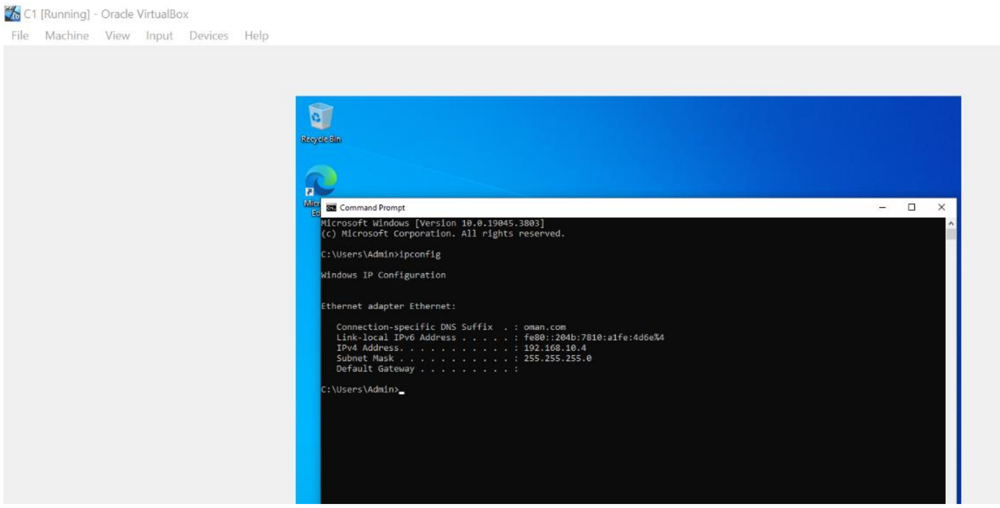

ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
DNS Manager
File Action View Help
DNS
ZAHRA-S1
Forward Lookup Zones
_msdcs.oman.com
MyprimaryZone
om
edu
oman.com
Reverse Lookup Zones
2.168.192.in-addr.arp
Trust Points
Conditional Forwarders
Name
Type
Data
Timestamp
om
(same as parent folder)
Start of Authority (SOA)
[1], zahra-s1.oman.com., ...
static
(same as parent folder)
Name Server (NS)
zahra-s1.oman.com.
static
X
Host (A)
192.168.10.2
Events
Services
Performance
BPA results
8:41 PM
12/4/2025
Right Ctrl

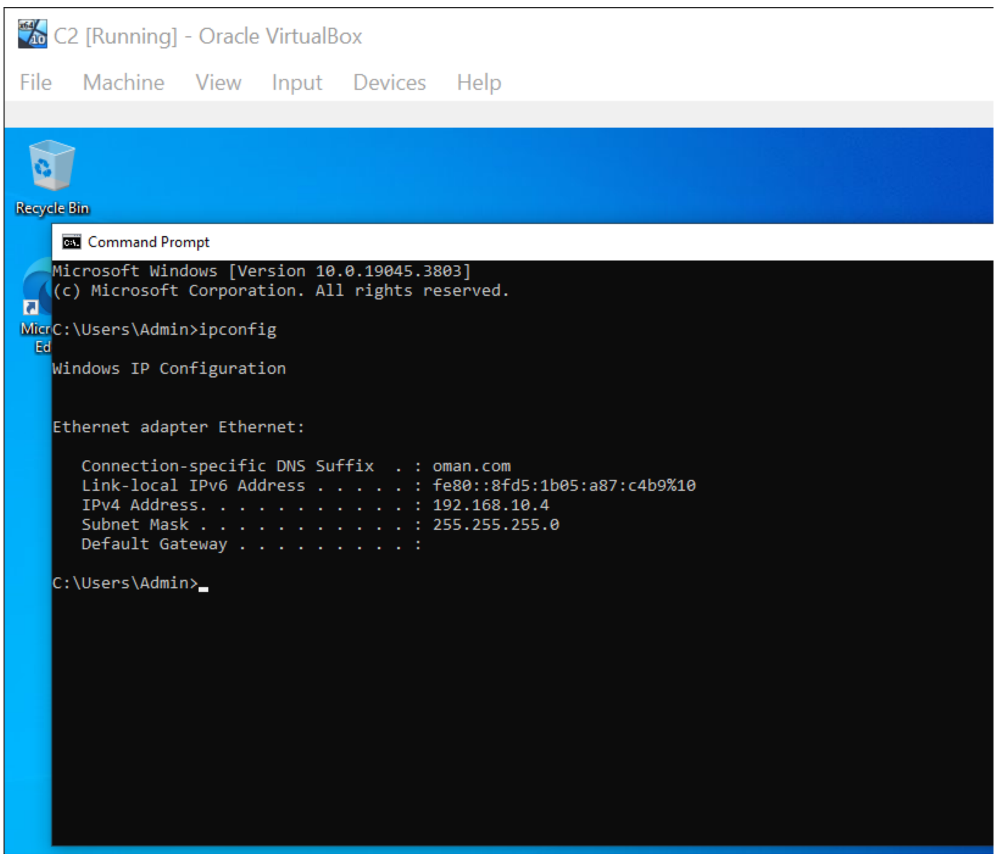
SERVER1:


ZAHRA-S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Administrator: Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Users\Administrator>ipconfig/renew
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : oman.com
Link-local IPv6 Address . . . . . : fe80::6451:55e:81e1:6d1e%4
IPv4 Address. . . . . : 192.168.10.3
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . :
Tunnel adapter isatap.oman.com:
Media State . . . . : Media disconnected
Connection-specific DNS Suffix . : oman.com
C:\Users\Administrator>
1:01 AM
12/5/2025
Right Ctrl


ZAHRA-S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Add Roles and Features Wizard
Select server roles
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
Confirmation
Select one or more roles to install on the selected server.
Roles
Active Directory Certificate Services
Active Directory Domain Services
Active Directory Federation Services
Active Directory Lightweight Directory Services
Active Directory Rights Management Services
Device Health Attestation
DHCP Server
DNS Server
Fax Server
File and Storage Services (1 of 12 installed)
Host Guardian Service
Hyper-V
MultiPoint Services
Network Policy and Access Services
Print and Document Services
Remote Access
Remote Desktop Services
Volume Activation Services
Web Server (IIS)
Windows Deployment Services
Description
Active Directory Domain Services (AD DS) stores information about objects on the network and makes this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.
< Previous
Next >
Install
Cancel
Events
Performance
BPA results
Services
Right Ctrl

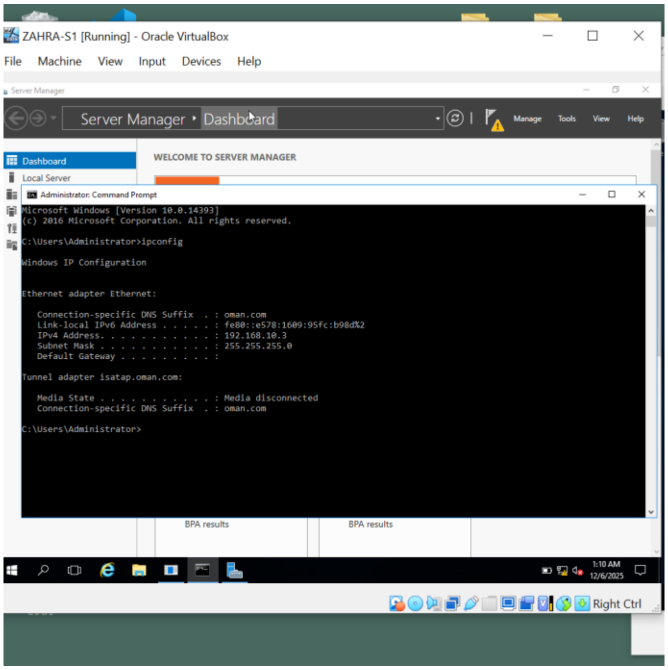

ZAHRA-S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Server Manager • Dashboard
Manage Tools View Help
Dashboard
AD DS
Add Roles and Features Wizard
Installation progress
View installation progress
Feature installation
DHCP Server
Remote Server Administration Tools
Role Administration Tools
DHCP Server Tools
You can close this wizard without interrupting running tasks. View task progress or open this page again by clicking Notifications in the command bar, and then Task Details.
Export configuration settings
< Previous
Next >
Close
Cancel
BPA results
Right Ctrl

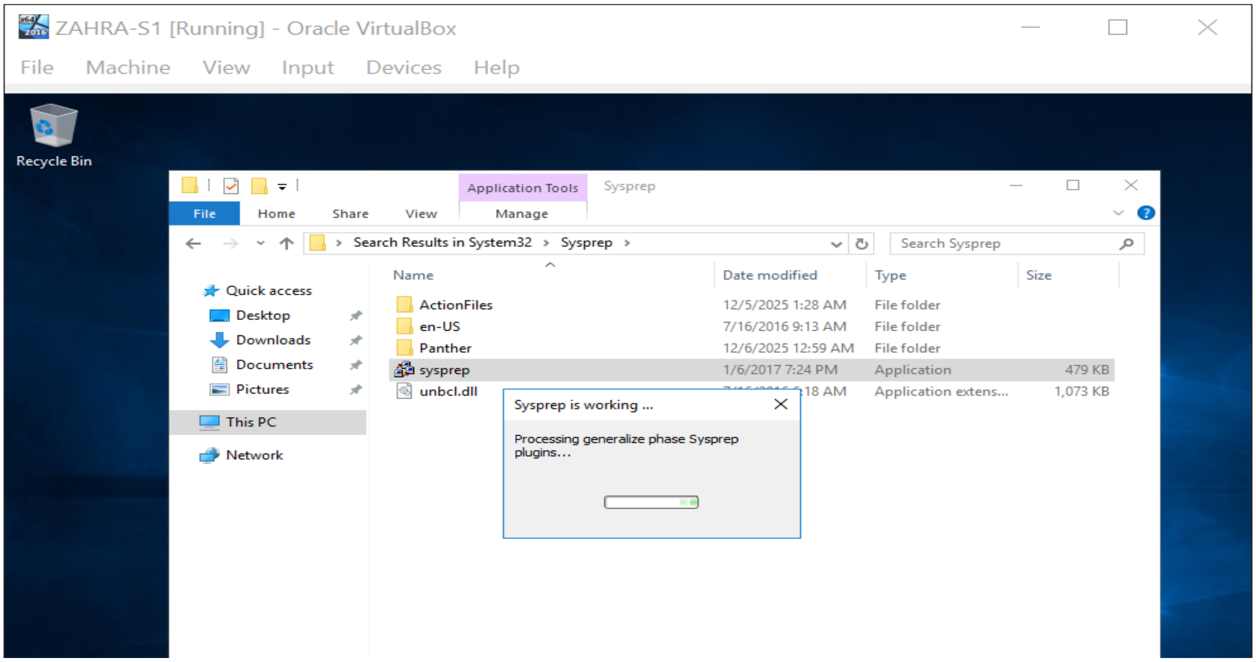

C1 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Recycle Bin
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Admin>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : oman.com
Link-local IPv6 Address . . . . . : fe80::204b:7810:a1fe:4d6e%4
IPv4 Address. . . . . : 192.168.10.4
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . :
C:\Users\Admin>

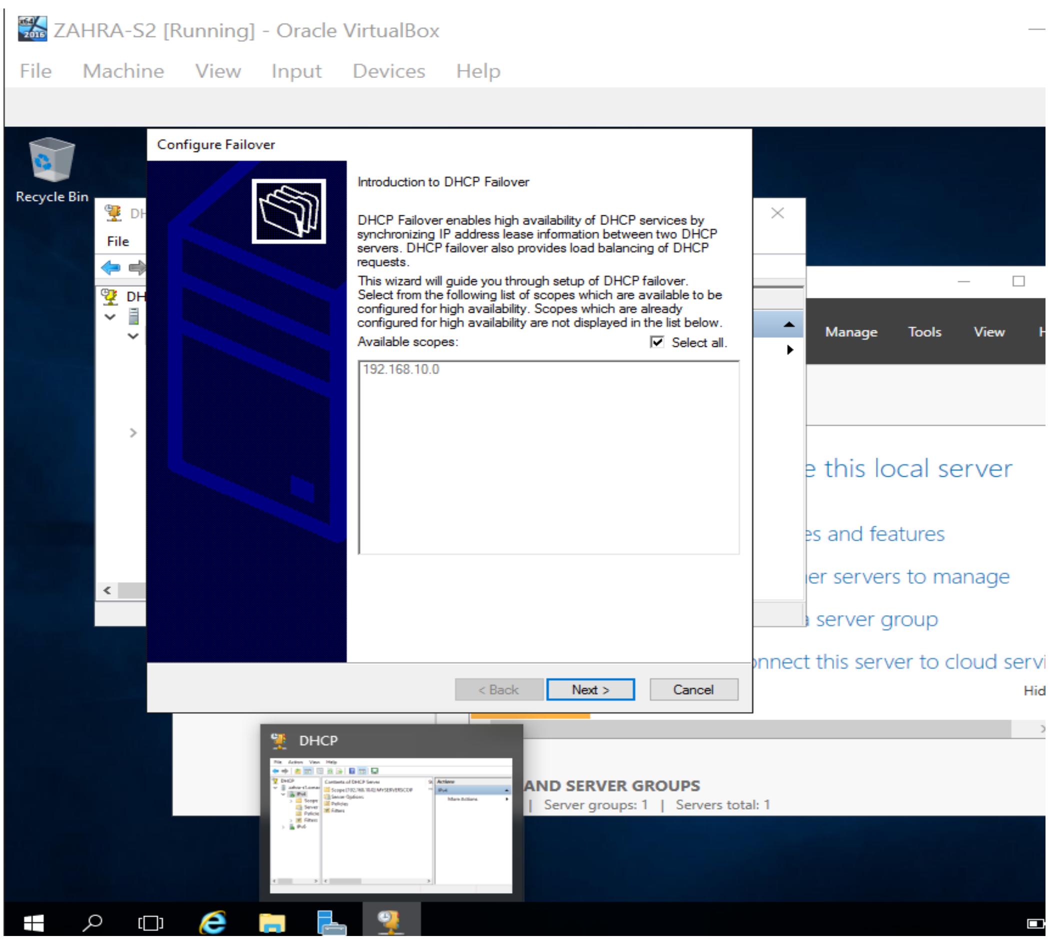

C2 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Recycle Bin
Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Admin>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : oman.com
Link-local IPv6 Address . . . . . : fe80::8fd5:1b05:a87:c4b9%10
IPv4 Address. . . . . : 192.168.10.4
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . :
C:\Users\Admin>

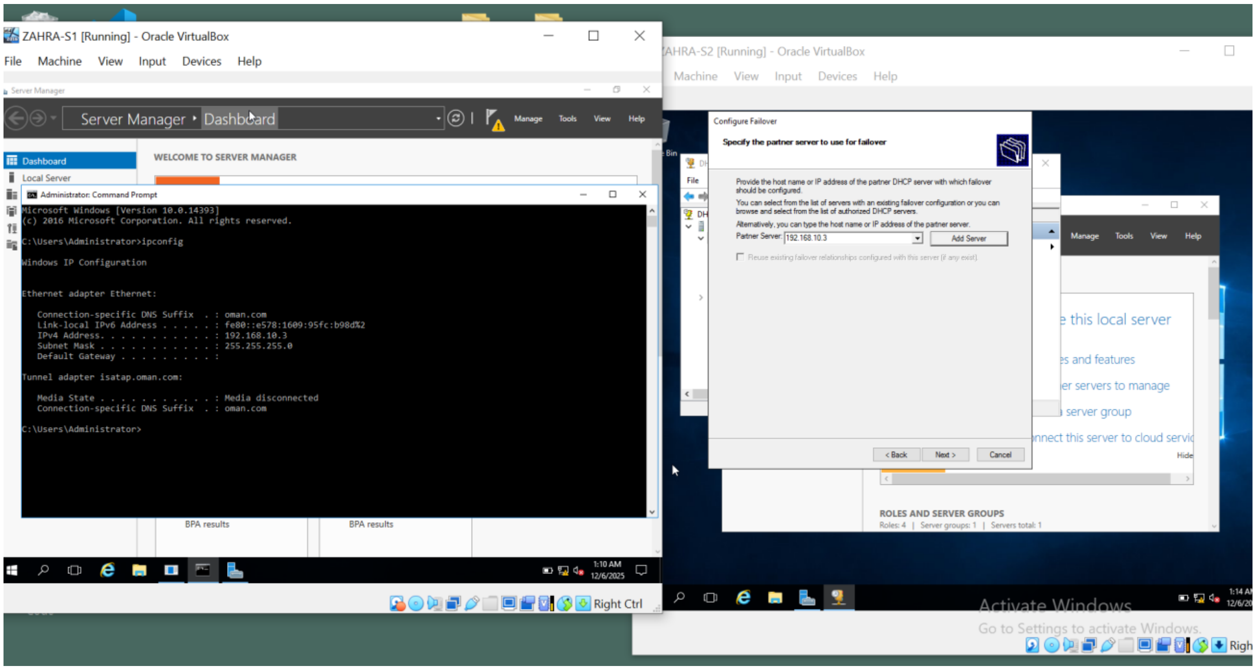
16.Configure YourName-S-2 with roles ADDC, DNS (don’t configure just attach with oman.com), DHCP (just add no configuration).

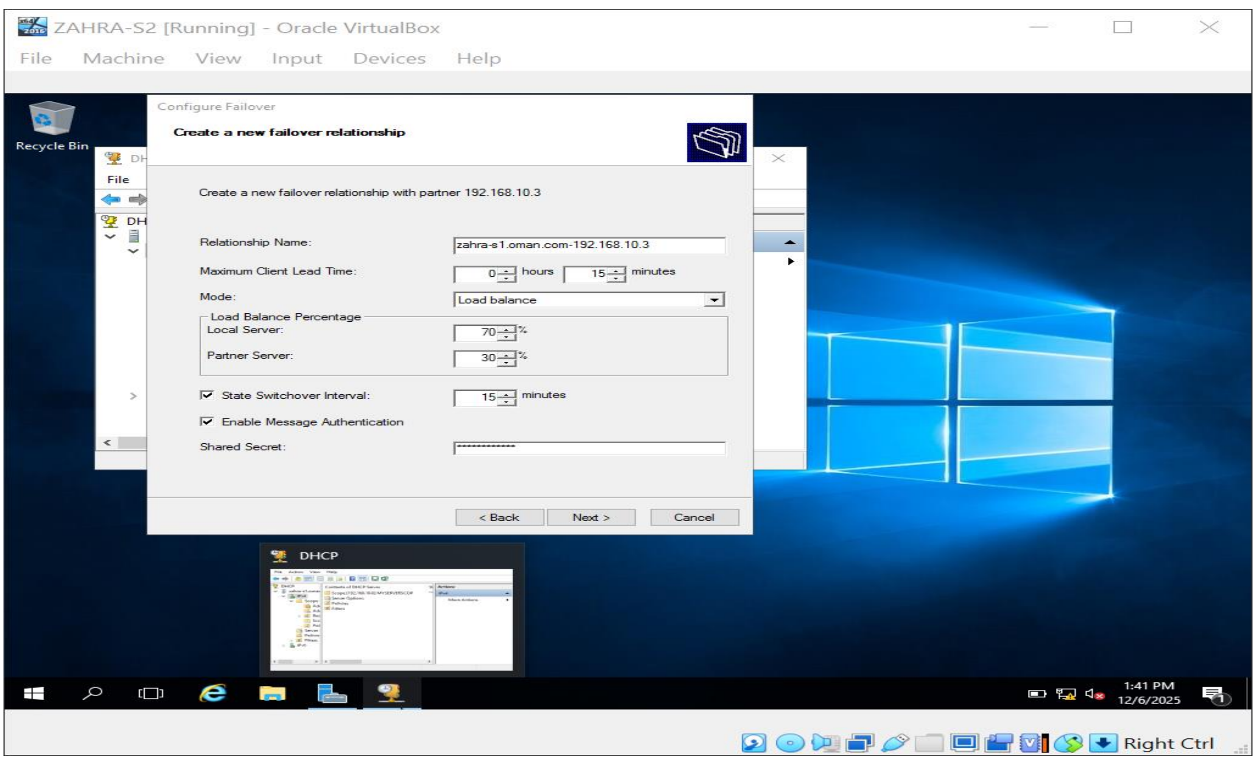
17. Install DHCP Failover server on YourName-S-1

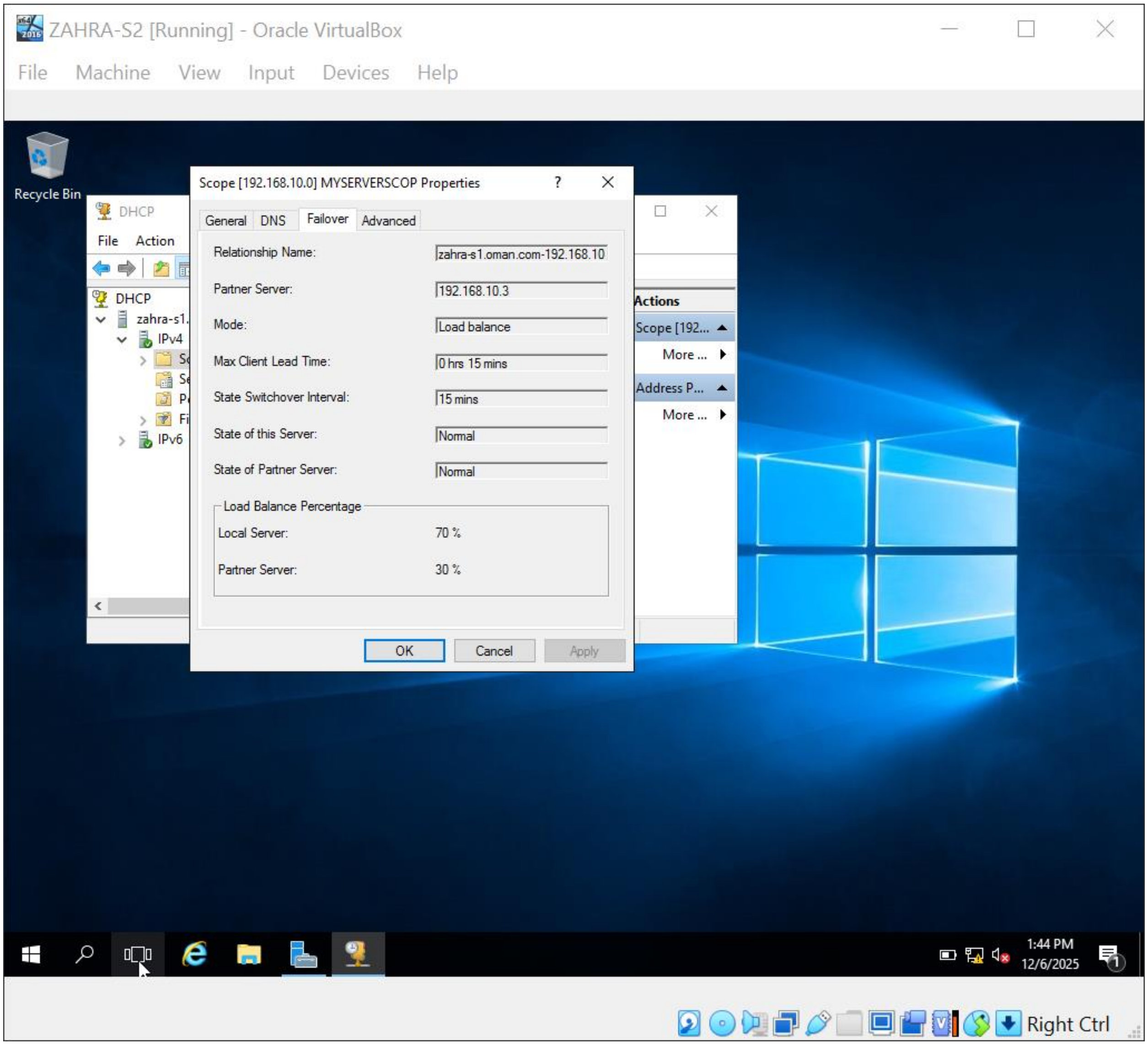



ZAHRA-S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Application Tools
Sysprep
File Home Share View Manage
Search Results in System32 > Sysprep >
Search Sysprep
Quick access
Desktop
Downloads
Documents
Pictures
This PC
Network
Name Date modified Type Size
ActionFiles 12/5/2025 1:28 AM File folder
en-US 7/16/2016 9:13 AM File folder
Panther 12/6/2025 12:59 AM File folder
sysprep 1/6/2017 7:24 PM Application 479 KB
unbcl.dll Application extens... 1,073 KB
Sysprep is working ...
Processing generalize phase Sysprep plugins...

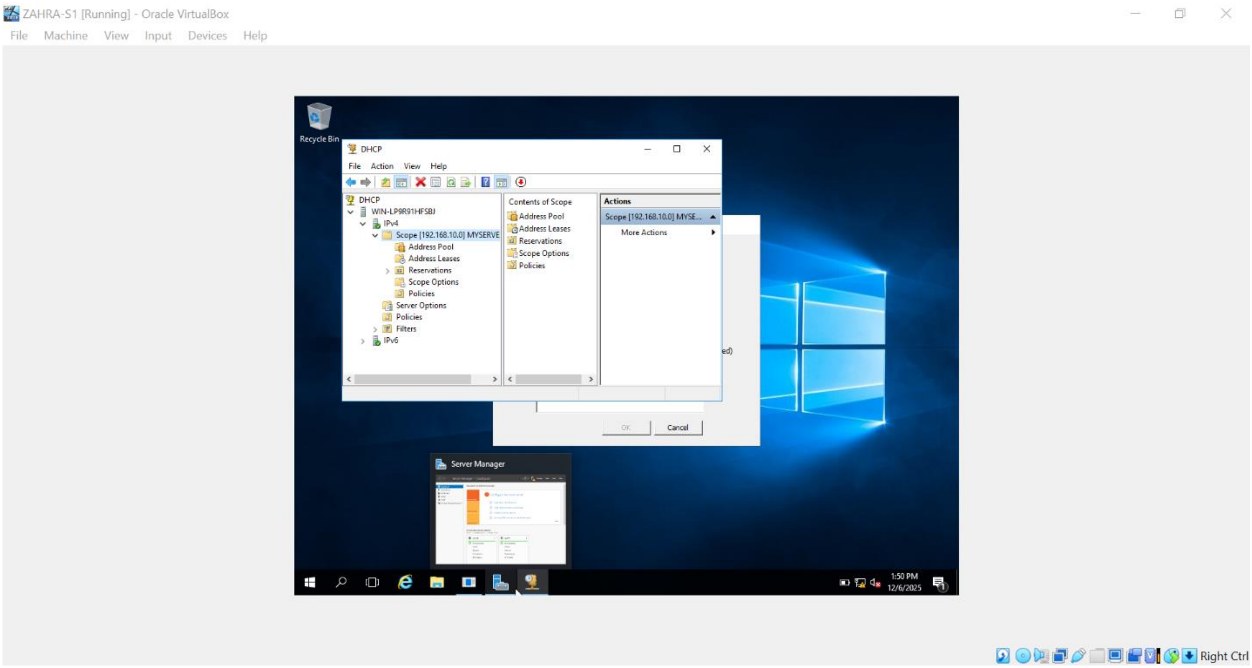

ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Configure Failover
Introduction to DHCP Failover
DHCP Failover enables high availability of DHCP services by synchronizing IP address lease information between two DHCP servers. DHCP failover also provides load balancing of DHCP requests.
This wizard will guide you through setup of DHCP failover. Select from the following list of scopes which are available to be configured for high availability. Scopes which are already configured for high availability are not displayed in the list below.
Available scopes:
Select all.
192.168.10.0
< Back
Next >
Cancel
Manage Tools View
e this local server
es and features
er servers to manage
server group
nnect this server to cloud servi
AND SERVER GROUPS
Server groups: 1 | Servers total: 1
DHCP

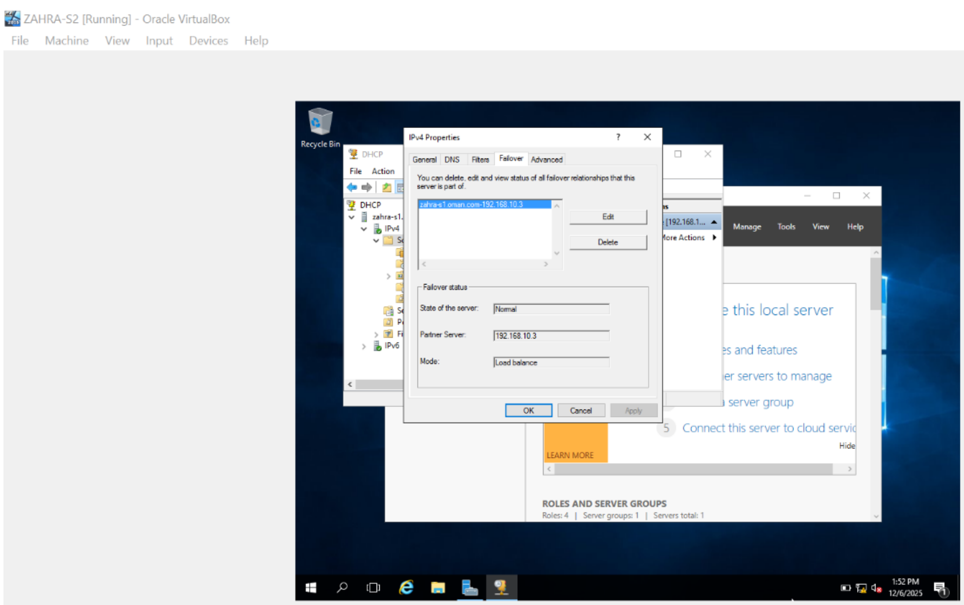

ZAHRA-S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Server Manager • Dashboard
Manage Tools View Help
Dashboard
Local Server
WELCOME TO SERVER MANAGER
Administrator: Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.
C:\Users\Administrator>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . : oman.com
Link-local IPv6 Address . . . . . : fe80::e578:1609:95fc:b98d%2
IPv4 Address. . . . . : 192.168.10.3
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . :
Tunnel adapter isatap.oman.com:
Media State . . . . : Media disconnected
Connection-specific DNS Suffix . : oman.com
C:\Users\Administrator>
BPA results
BPA results
1:10 AM
12/6/2025
Right Ctrl
ZAHRA-S2 [Running] - Oracle VirtualBox
Machine View Input Devices Help
Configure Failover
Specify the partner server to use for failover
Provide the host name or IP address of the partner DHCP server with which failover should be configured.
You can select from the list of servers with an existing failover configuration or you can browse and select from the list of authorized DHCP servers.
Alternatively, you can type the host name or IP address of the partner server.
Partner Server: 192.168.10.3
Add Server
Reuse existing failover relationships configured with this server (if any exist)
< Back
Next >
Cancel
this local server
s and features
er servers to manage
server group
nnect this server to cloud servic
ROLES AND SERVER GROUPS
Roles: 4 | Server groups: 1 | Servers total: 1
Activate Windows
Go to Settings to activate Windows.


ZAHRA-S2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Configure Failover
Create a new failover relationship
Create a new failover relationship with partner 192.168.10.3
Relationship Name: zahra-s1.oman.com-192.168.10.3
Maximum Client Lead Time: 0 hours 15 minutes
Mode: Load balance
Load Balance Percentage
Local Server: 70 %
Partner Server: 30 %
State Switchover Interval: 15 minutes
Enable Message Authentication
Shared Secret:
< Back
Next >
Cancel
DHCP
1:41 PM
12/6/2025
Right Ctrl


ZAHRA-S2 [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Recycle Bin
DHCP
File
Action
DHCP
zahra-s1.
IPv4
IPv6
Scope [192.168.10.0] MYSERVERSCOP Properties
General
DNS
Failover
Advanced
Relationship Name:
zahra-s1.oman.com-192.168.10
Partner Server:
192.168.10.3
Mode:
Load balance
Max Client Lead Time:
0 hrs 15 mins
State Switchover Interval:
15 mins
State of this Server:
Normal
State of Partner Server:
Normal
Load Balance Percentage
Local Server:
70 %
Partner Server:
30 %
OK
Cancel
Apply
Actions
Scope [192...
More ...
Address P...
More ...
1:44 PM
12/6/2025
Right Ctrl

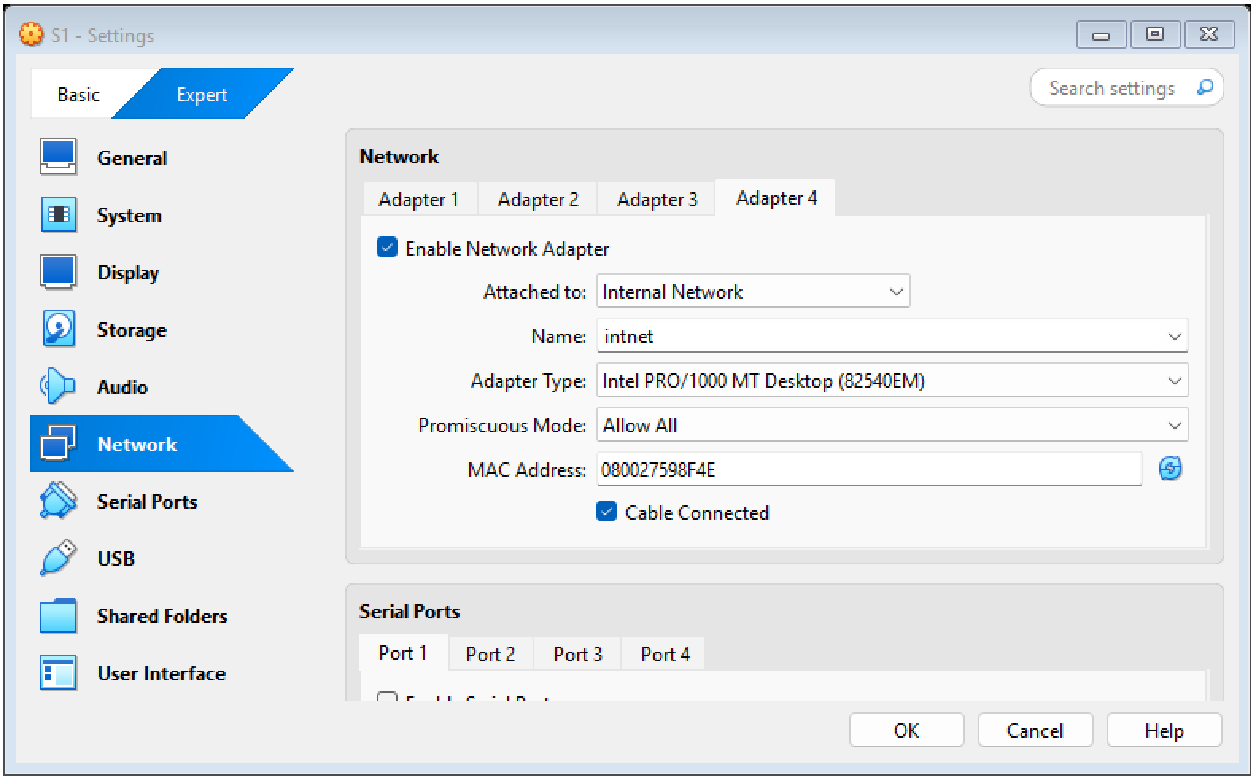
18. YourName-S-1 Connecting to YourName-S-2. Show Test.

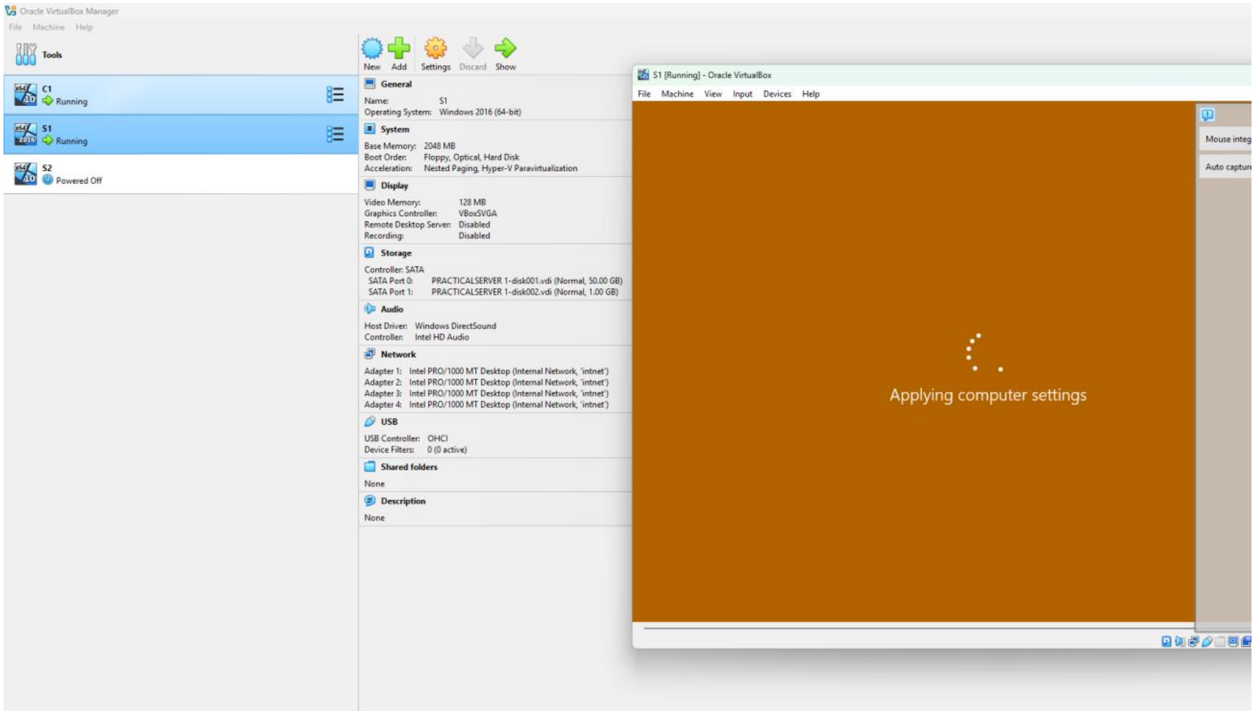


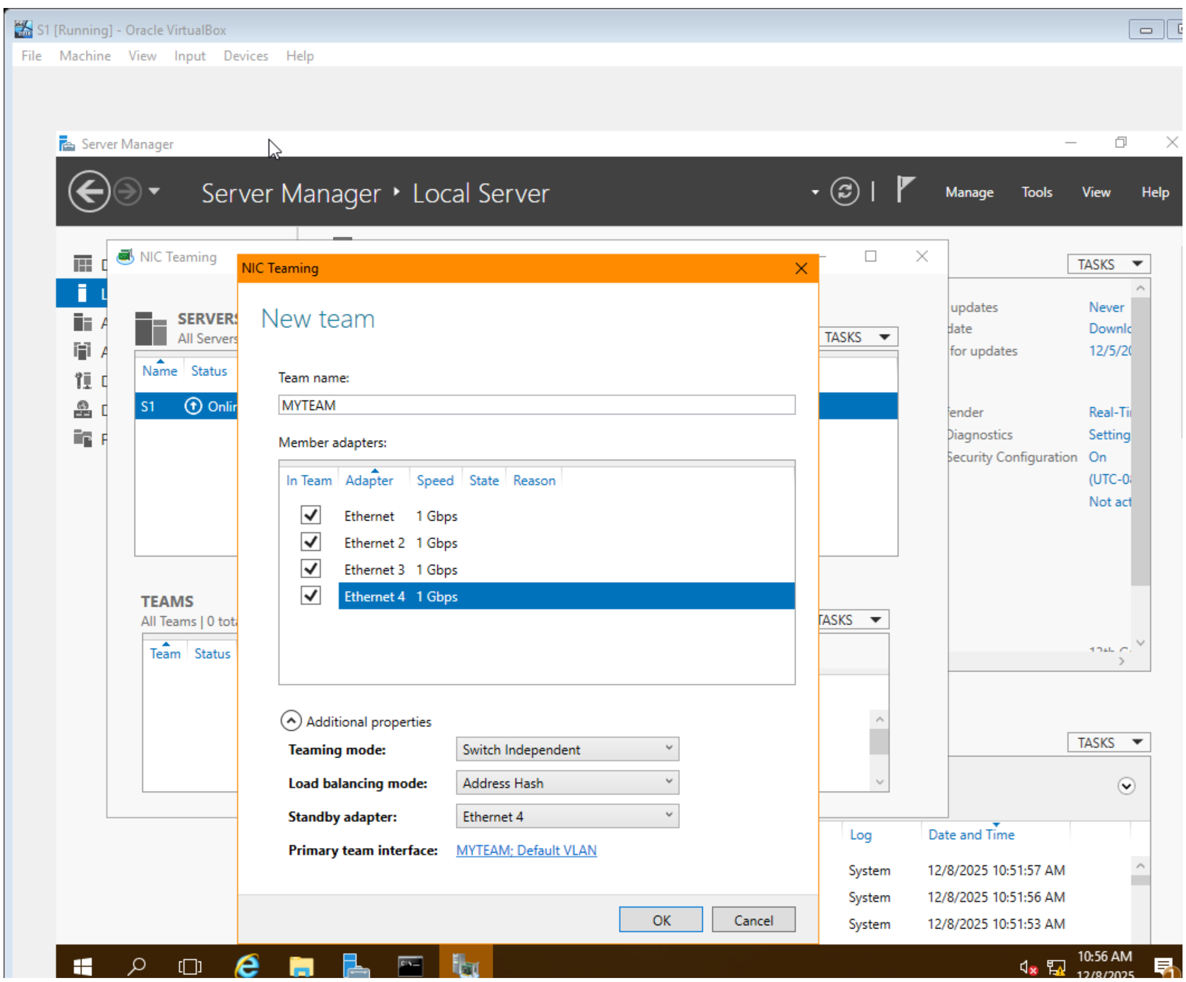


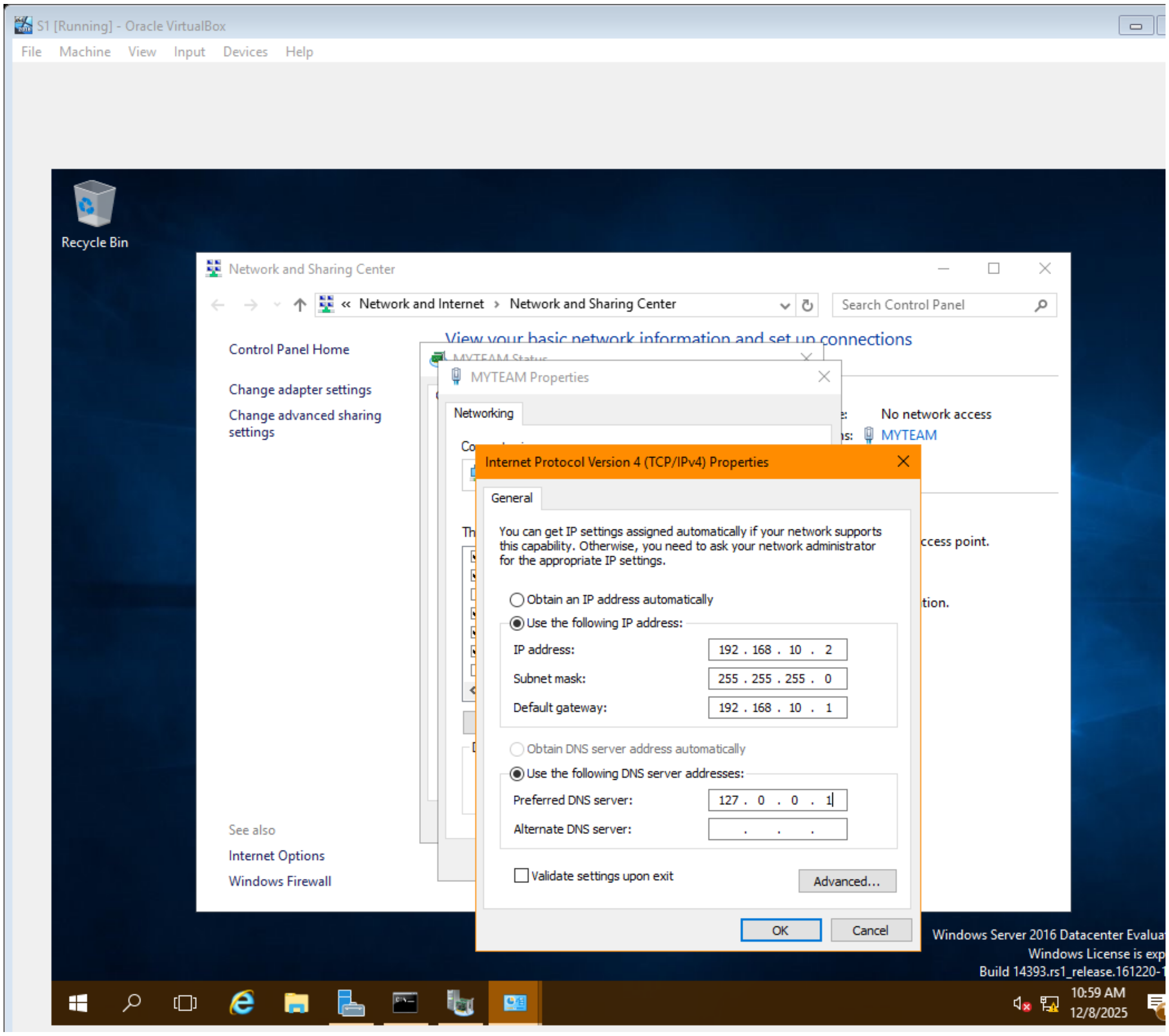
19.Configuring NIC Teaming, by adding adapters.

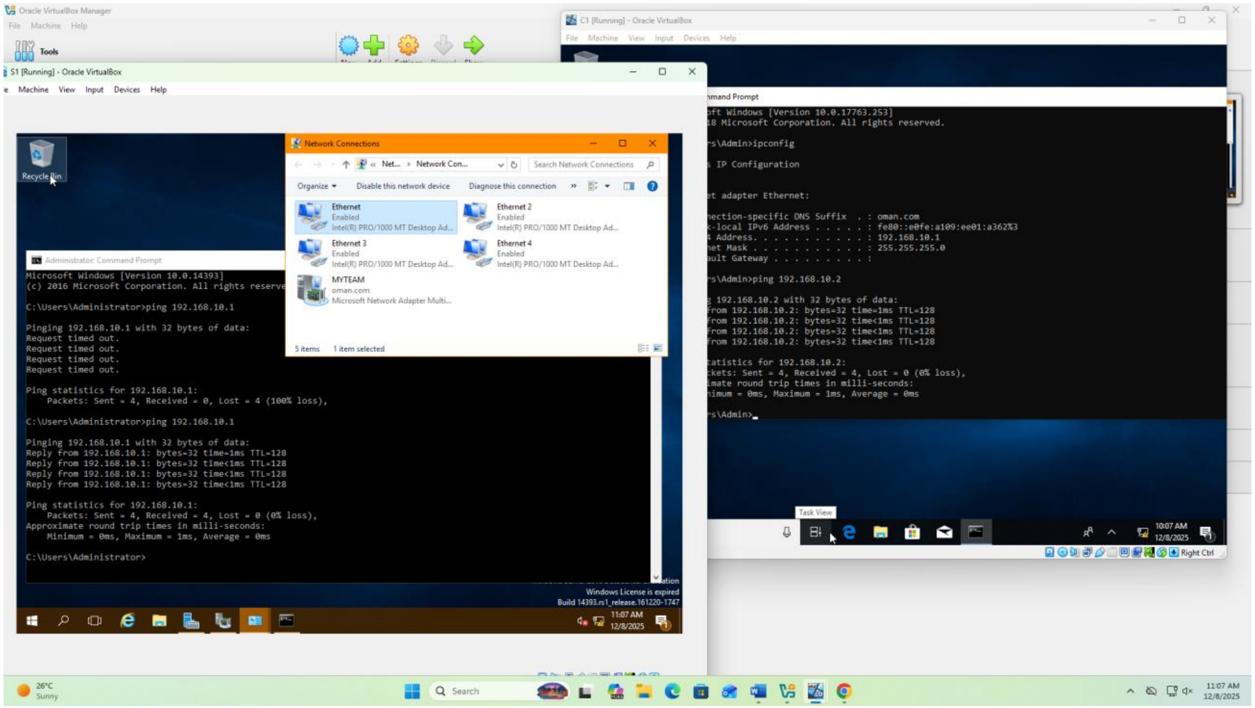
20. Disable adapters in NIC Teaming and show connectivity results.

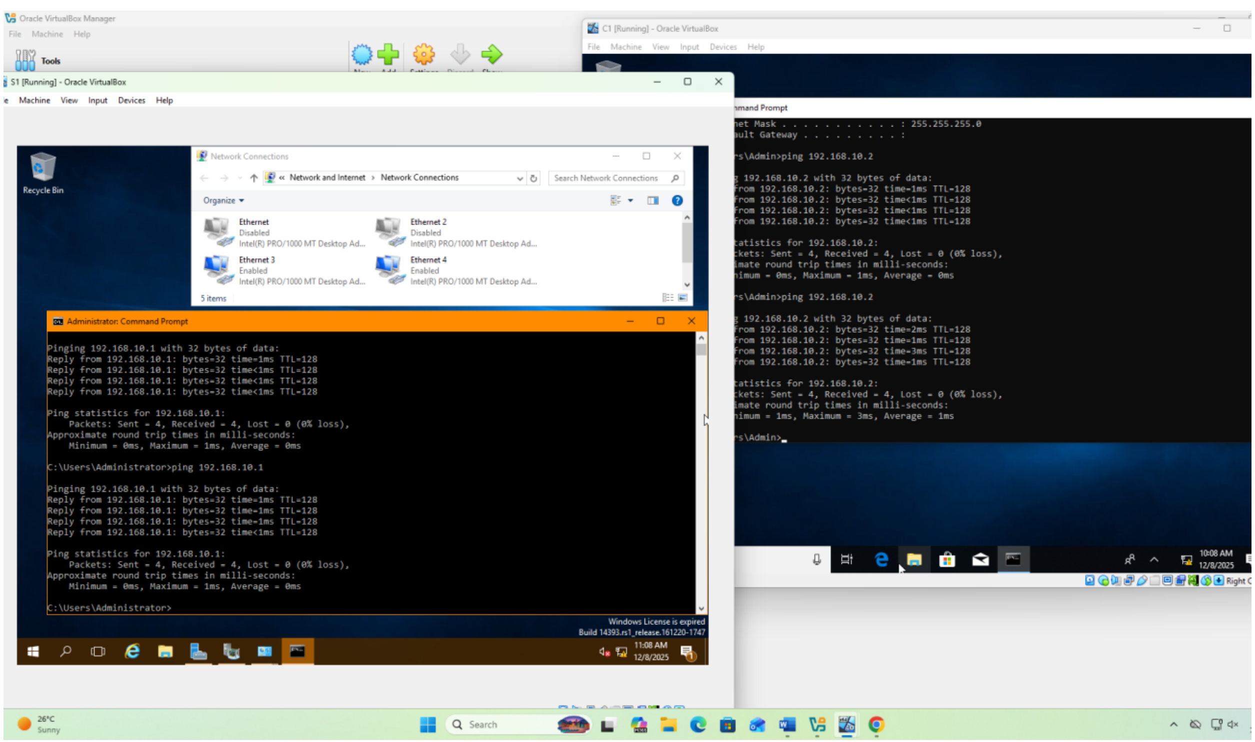



Oracle VirtualBox Manager
File Machine Help
Tools
C1 Running
S1 Running
S2 Powered Off
New Add Settings Discard Show
General
Name: S1
Operating System: Windows 2016 (64-bit)
System
Base Memory: 2048 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: PRACTICALSERVER 1-disk001.vdi (Normal, 50.00 GB)
SATA Port 1: PRACTICALSERVER 1-disk002.vdi (Normal, 1.00 GB)
Audio
Host Driver: Windows DirectSound
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (Internal Network, 'intnet')
Adapter 2: Intel PRO/1000 MT Desktop (Internal Network, 'intnet')
Adapter 3: Intel PRO/1000 MT Desktop (Internal Network, 'intnet')
Adapter 4: Intel PRO/1000 MT Desktop (Internal Network, 'intnet')
USB
USB Controller: OHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None
S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Applying computer settings


S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Server Manager
Server Manager ▸ Local Server
Manage Tools View Help
NIC Teaming
SERVERS
All Servers
Name Status
S1 Online
TEAMS
All Teams | 0 total
Team Status
TASKS
NIC Teaming
New team
Team name:
MYTEAM
Member adapters:
In Team Adapter Speed State Reason
Ethernet 1 Gbps
Ethernet 2 1 Gbps
Ethernet 3 1 Gbps
Ethernet 4 1 Gbps
Additional properties
Teaming mode: Switch Independent
Load balancing mode: Address Hash
Standby adapter: Ethernet 4
Primary team interface: MYTEAM: Default VLAN
OK Cancel
updates Never
for updates
Security Configuration On
Log Date and Time
System 12/8/2025 10:51:57 AM
System 12/8/2025 10:51:56 AM
System 12/8/2025 10:51:53 AM
10:56 AM


S1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Network and Sharing Center
Network and Internet > Network and Sharing Center
Search Control Panel
Control Panel Home
Change adapter settings
Change advanced sharing settings
View your basic network information and set up connections
MYTEAM Properties
Networking
No network access
MYTEAM
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 192 . 168 . 10 . 2
Subnet mask: 255 . 255 . 255 . 0
Default gateway: 192 . 168 . 10 . 1
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server: 127 . 0 . 0 . 1
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
See also
Internet Options
Windows Firewall
Windows Server 2016 Datacenter Evalua
Windows License is exp
Build 14393.rs1_release.161220-1
10:59 AM
12/8/2025


S1 [Running] - Oracle VirtualBox
C1 [Running] - Oracle VirtualBox
Network Connections
Ethernet
Enabled
Ethernet 2
Enabled
Ethernet 3
Enabled
Ethernet 4
Enabled
MYTEAM
oman.com
Microsoft Network Adapter Multi...
Microsoft Windows [Version 10.0.14393]
C:\Users\Administrator>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Users\Administrator>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\Administrator>
IP Configuration
Connection-specific DNS Suffix . : oman.com
Address. . . . . : 192.168.10.1
Mask . . . . : 255.255.255.0
ping 192.168.10.2
192.168.10.2 with 32 bytes of data:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),


S1 [Running] - Oracle VirtualBox
C1 [Running] - Oracle VirtualBox
Network Connections
Network and Internet › Network Connections
Ethernet
Disabled
Ethernet 2
Disabled
Ethernet 3
Enabled
Ethernet 4
Enabled
Administrator: Command Prompt
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\Administrator>ping 192.168.10.1
ping 192.168.10.2
192.168.10.2 with 32 bytes of data:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),